surface layer (the most flocculent and highly contaminated yeasts) is removed and discarded and the underlying cells (the 'middle skimmings') are harvested and used for subsequent pitching. Therefore, the 'middle skimmings' contain cells which have the desired flocculence and which have been protected from contamination by the surface layer of the yeast head. The pitching yeast may be treated to reduce the level of contaminating bacteria and remove protein and dead yeast cells by such treatments as reducing the pH of the slurry to 2.5 to 3, washing with water, washing with ammonium persulphate and treatment with antibiotics such as polymixin, penicillin and neomycin (Mandl *et al.,* 1964; Strandskov, 1964; Roessler, 1968; Reed and Nagodawithana, 1991a).

However, traditional open vessels are becoming increasingly rare and the bulk of beer is brewed using cylindro-conical fermenters (see Chapter 7). In these systems the yeast flocculates and collects in the cone at the bottom of the fermenter where it is subject to the stresses of nutrient starvation, high ethanol concentration, low water activity, high carbon dioxide concentration and high pressure (Boulton, 1991). Thus, the viability and physiological state of the yeast crop would not be ideal for an inoculum. The viability of the crop may be assessed using a biomass probe of the type described earlier, thus ensuring that at least the correct amount of viable biomass is used to start the next fermentation. However, the physiological state of the biomass will not have been influenced by such monitoring procedures. The situation is further complicated by the fact that the harvested yeast is stored before it is used as inoculum. Metabolic activity is minimized during this time by chilling rapidly to about 1°, suspending in beer and storing in the absence of oxygen. If oxygen is present during the storage period then the yeast cells consume their stored glycogen which renders them very much less active at the start of the fermentation (Pickerell *et al.,* 1991).

One of the key physiological features of yeast inoculum is the level of sterol in the cells. Sterols are required for membrane synthesis but they are only produced in the presence of oxygen. Thus, we have the anomaly of oxygen being required for sterol synthesis, yet anaerobic conditions are required for ethanol production. This anomaly is resolved traditionally by aerating the wort before inoculation. This oxygen allows sufficient sterol synthesis early in the fermentation to support growth of the cells throughout the process, that is after the oxygen is exhausted and the process is anaerobic. Boulton *et al.* (1991) developed an alternative approach where the pitching yeast was vigorously aerated prior to inoculation. The yeast was then sterol rich and had no requirement for oxygen during the alcohol fermentation.

The difficulties outlined above and the likelihood of strain degeneration and contamination mean that yeasts are rarely used for more than five to ten consecutive fermentations (Thorne, 1970; Reed and Nagodawithana, 1991a) which necessitates the periodical production of a pure inoculum. This would involve developing sufficient biomass from a single colony to pitch a fermentation at a level of approximately 2 grams of pressed yeast per litre. Hansen (1896) pioneered the use of pure inocula and devised a yeast propagation scheme utilizing a 10% inoculum volume at each stage in the programme and employing conditions similar to those used during brewing. However, modern propagation schemes use inoculum volumes of 1% or even lower and may use conditions different from those used during brewing. Therefore, continuous aeration may be used during the propagation stage which seems to have little effect on the beer produced in the subsequent fermentation (Curtis and Clark, 1957). Yeast inoculum produced in this way would also be sterol rich, obviating the need for aerated wort.

A number of yeast propagators (which are basically closed, aerated vessels) have been described in the literature. The simplest type of progagator is a single-stage system resembling an unstirred, aerated fermenter which is inoculated with a shake-flask culture developed from a single colony (Gilliland, 1971). Curtis and Clark (1957) and Thorne (1970) described two-stage systems which could be operated semi-continuously. Thorne's propagator consisted of two linked vessels, 1.5 and 150 dm^3 respectively. The smaller vessel was filled with wort, sterilized, cooled, aerated and inoculated with a flask-grown culture. After growth for 3 to 4 days the culture was forced by air pressure into the second vessel which had been filled with sterilized, cooled wort and aerated. An aliquot of 1.5 dm^3 was forced back into the first vessel after mixing. In a further 3 to 4 days the larger vessel contained sufficient biomass to pitch a 1000 dm^3 fermenter and the first vessel contained sufficient inoculum for another second stage. However, although this procedure should produce a pure inoculum there is a danger of strain degeneration occurring in such a semi-continuous system.

Boulton (1991) speculated that when the UK moves to an alcohol-based tax then the economics of propagating inoculum for each brew would be considerably more attractive. He suggested that the bakers' yeast aerobic fed-batch inoculum programme could be adopted to produce sterol-rich catabolite-derepressed cells. Such an inoculum would remove the necessity for aerating the wort prior to inoculation because sterol synthesis would not be necessary due to the high sterol content of the cells.

Bakers' yeast

The commercial production of bakers' yeast involves the development of an inoculum through a large number of aerobic stages. Although the production stages of the process may not be operated under strictly aseptic conditions a pure culture is used for the initial inoculum, thereby keeping contamination to a minimum in the early stages of growth. Reed and Nagodawithana (1991b) discussed the development of inoculum for the production of bakers' yeast and quoted a process involving eight stages, the first three being aseptic while the remaining stages were carried out in open vessels. The yeast may be pumped from one stage to the next or the seed cultures may be centrifuged and washed before transfer, which reduces the level of contamination. The yields obtained in the first five stages are relatively low because they are not fed-batch systems, whereas the last three stages are fed-batch (the fed-batch bakers' yeast fermentation is considered in more detail in Chapter 2). A summary of a typical inoculum development programme for the production of bakers' yeast is given in Fig. 6.2.

THE DEVELOPMENT OF INOCULA FOR BACTERIAL PROCESSES

The main objective of inoculum development for traditional bacterial fermentations is to produce an active inoculum which will give as short a lag phase as possible in subsequent culture. A long lag phase is disadvantageous in that not only is time wasted but also medium is consumed in maintaining a viable culture prior to growth. The length of the lag phase is affected by the size of the inoculum and its physiological condition (Meyrath and Suchanek, 1972). As already stated, the inoculum size normally ranges between 3 and 10% of the culture volume. Lincoln (1960) stressed that bacterial inocula should be transferred in the logarithmic phase of growth, when the cells are still metabolically active. The age of the inoculum is particularly important in the growth of sporulating bacteria, for sporulation is induced at the end of the logarithmic phase and the use of an inoculum containing a high percentage of spores would result in a long lag phase in a subsequent fermentation.

Keay *et al.* (1972) quote the use of a 5% inoculum of a logarithmically growing culture of a thermophilic *Bacillus* for the production of proteases. Aunstrup (1974) described a two-stage inoculum development programme for the production of proteases by *Bacillus subtilis*. Inoculum for a seed fermenter was grown for 1

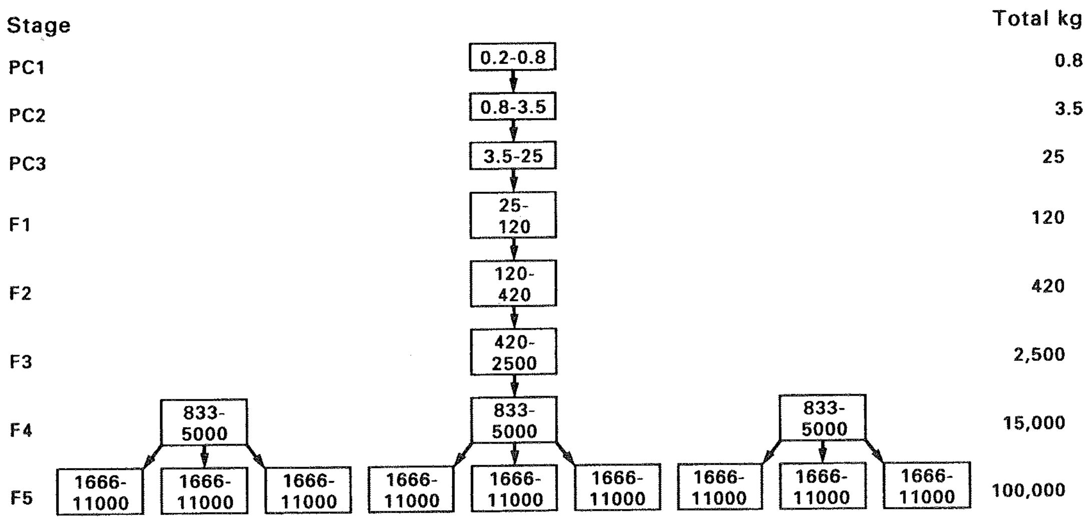

FIG. 6.2. The development of inoculum for the commercial production of bakers' yeast. PC 1, 2 and 3 are pure culture batch fermentations. F1 and 2 are non-aseptic batch fermentations. F3 and 4 are fed-batch fermentations and F5 is the final fed-batch fermentation (Reed and Nagodawithana, 1991a).

to 2 days on a solid or liquid medium and then transferred to a seed vessel where the organism was allowed to grow for a further ten generations before transfer to the production stage. Priest and Sharp (1989) cited the use of a 5% inoculum, still in the exponential phase, for the commercial production of *Bacillus* α-amylase. Underkofler (1976) emphasized that, in the production of bacterial enzymes, the lag phase in plant fermenters could be almost completely eliminated by using inoculum medium of the same composition as used in the production fermenter and employing large inocula of actively growing seed cultures. The inoculum development programme for a pilot-plant scale process for the production of vitamin B_{12} from *Pseudomonas denitrificans* is shown in Fig. 6.3 (Spalla *et al.*, 1989).

The necessity to use an inoculum in an active physiological state is taken to its extreme in the production of vinegar. The acetic-acid bacteria used in the vinegar process are extremely sensitive to oxygen starvation. Therefore, to avoid disturbing the system, the cells at the end of a fermentation are used as inoculum for the next batch by removing approximately 60% of the culture and restoring the original level with fresh medium (Conner and Allgeier, 1976). The advantage of a highly active inoculum apparently outweighs the disadvantages of possible strain degeneration and contaminant accumulation. However, strain stability is a major concern in inoculum development for fermentations employing recombinant bacteria. Sabatie *et al.* (1991) demonstrated that plasmid stability and productivity in an *E. coli* biotin fermentation was greatly improved if stationary, rather then exponential, phase cells were used as inoculum. They postulated that the plasmid copy number may be higher in stationary cells than in exponential ones, resulting in a lower plasmid loss in the subsequent fermentation when a stationary culture is used as inoculum. A stationary phase inoculum would result in a lag phase, but this disadvantage was more than compensated for by the considerable improvement in plasmid retention and biotin production compared with that obtained using an exponential inoculum.

In the lactic-acid fermentation the producing organism may be inhibited by lactic acid. Thus, production of lactic acid in the seed fermentation may result in the generation of poor quality inoculum. Yamamoto *et al.* (1993) generated high quality inoculum of *Lactococcus lactis* IO-1 on a laboratory scale using electrodialysis seed culture which reduced the lactate in the inoculum and reduced the length of the lag phase in the production fermentation.

An example of the development of inoculum for an anaerobic bacterial process is provided by the clostridial acetone–butanol fermentation. The process was outcompeted by the petrochemical industry but there is still considerable interest in re-establishing the fermen-

STOCK CULTURE

Lyophilised with skim milk

↓

MAINTENANCE CULTURE

Agar slope incubated 4 days at 28°

↓

SEED CULTURE - FIRST STAGE

2 dm^3 flask containing 0.6 dm^3 medium inoculated with culture from one slope; incubated with shaking for 48h at 28°

↓

SEED CULTURE - SECOND STAGE

40 - 80 dm^3 fermenter containing 25 - 50 dm^3 medium inoculated with 1 - 1.2% first stage seed culture. Incubated 25 - 30h at 32°

↓

PRODUCTION CULTURE

500 dm^3 fermenter with 300 dm^3 medium inoculated with 5% second stage seed culture. Incubated at 32° for 140 - 160 h

FIG. 6.3. The inoculum development programme for a vitamin B_{12} pilot scale fermentation using *Pseudomonas denitrificans* (Spalla *et al.*, 1989).

tation (Gottschalk and Grupe, 1992). The inoculum development programme described by McNeil and Kristiansen (1986) is given in Table 6.2. The stock culture is heat shocked to stimulate spore germination and to eliminate the weaker spores. The production stage is inoculated with a very low volume and this corresponds with Lurie's (1975) description of the South African acetone–butanol fermentation in which a 100,000 dm^3 fermenter was inoculated with only 10 dm^3 of seed. The use of such small inocula necessitates the achievement of as near perfect conditions as possible to prevent contamination and to avoid an abnormally long lag phase.

THE DEVELOPMENT OF INOCULA FOR MYCELIAL PROCESSES

The preparation of inocula for fermentations employing mycelial (filamentous) organisms is more involved than that for unicellular bacterial and yeast processes. The majority of industrially important fungi and streptomycetes are capable of asexual sporulation, so it is common practice to use a spore suspension as seed during an inoculum development programme. A major advantage of a spore inoculum is that is contains far more 'propagules' than a vegetative culture. Three basic techniques have been developed to produce a high concentration of spores for use as an inoculum.

Sporulation on solidified media

Most fungi and streptomycetes will sporulate on suitable agar media but a large surface area must be employed to produce sufficient spores. Parker (1950) described the 'roll-bottle' technique for the production of spores of *Penicillium chrysogenum*. Quantities of medium (300 cm^3) containing 3% agar were sterilized in 1 dm^3 cylindrical bottles, which were then cooled to 45° and rotated on a roller mill so that the agar set as a cylindrical shell inside the bottle. The bottles were inoculated with a spore suspension from a sub-master slope and incubated at 24° for 6 to 7 days. Parker claimed that although the use of the 'roll-bottle' involved some sacrifice in ease of visual examination, it provided a large surface area for cultivation of spores in a vessel of a convenient size for handling in the laboratory.

Hockenhull (1980) described the production of 10^{10} spores of *Penicillium chrysogenum* on a 300-cm^2 agar layer in a Roux bottle and El Sayed (1992) quoted the use of spore suspensions derived from agar media containing between 10^7 and 10^8 cm^{-3}. Butterworth (1984) described the use of a Roux bottle for the production of a spore inoculum of *Streptomyces clavuligerus* for the production of clavulanic acid. The spores produced from one bottle containing 200-cm^2 agar surface could be used to inoculate a 75-dm^3 seed fermenter which, in turn, was used to inoculate a 1500-dm^3 fermenter. The clavulanic acid inoculum development programme is illustrated in Fig. 6.4. Some representative solidified media for the production of streptomycete and fungal spores are given in Tables 6.3 and 6.4 respectively.

Sporulation on solid media

Many filamentous organisms will sporulate profusely on the surface of cereal grains from which the spores

TABLE 6.2. *The inoculum development programme for the clostridial acetone–butanol fermentation* (Spivey, 1978)

Stage	Cultural conditions	Incubation time (hours)
1.	Heat-shocked spore suspension inoculated into 150 cm^3 of potato glucose medium	12
2.	Stage 1 culture used as inoculum for 500 cm^3 molasses medium	6
3.	Stage 2 culture used as inoculum for 9 dm^3 molasses medium	9
4.	Stage 3 culture used as inoculum for 90,000 dm^3 molasses medium	

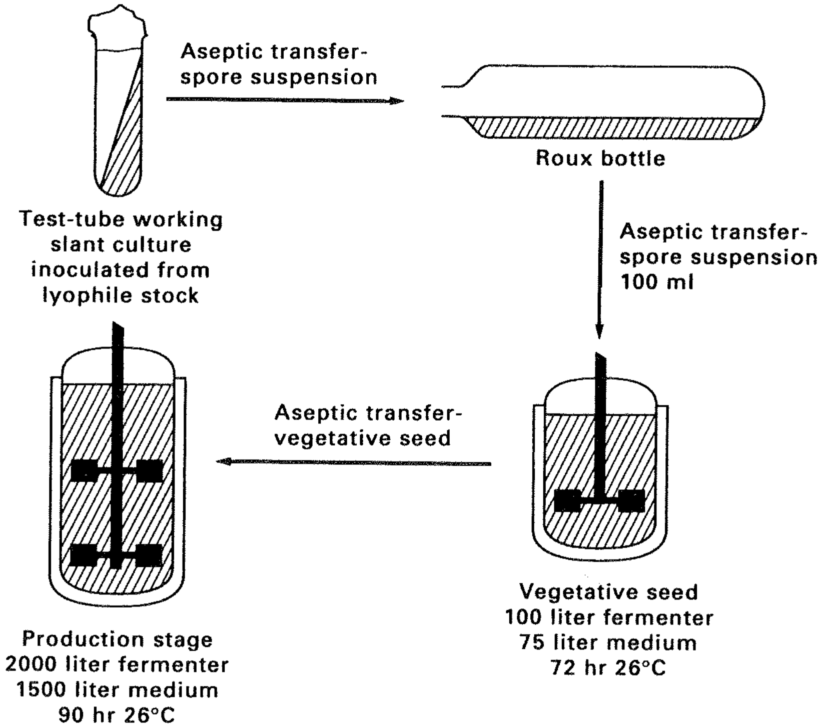

FIG. 6.4. The inoculum development programme for the production of clavulanic acid from *Streptomyces clavuligerus* (Butterworth, 1984).

may be harvested. Substrates such as barley, hard wheat bran, ground maize and rice are all suitable for the sporulation of a wide range of fungi. The sporulation of a given fungus is particularly affected by the amount of water added to the cereal before sterilization and the relative humidity of the atmosphere, which should be as high as possible during sporulation (Vezina and Singh, 1975). Singh *et al.* (1968) have described a system for the sporulation of *Aspergillus ochraces* in which a 2.8-dm^3 Fernbach flask containing 200 grams of 'pot' barley or 100 grams of moistened wheat bran produced 5×10^{11} conidia after six days at 28° and 98% relative humidity. This was 5 times the number obtainable from a Roux bottle batched with Sabouraud agar and 50 times the number obtainable from such a vessel batched with Difco Nutrient Agar, incubated for the same time period. Vezina *et al.* (1968) have published a list of fungi which are capable of sporulating heavily on cereal grains. El-Sayed (1992) quoted the use of cooked rice for the production of spores of *Penicillium* and *Cephalosporium* in penicillin and cephalosporin inoculum development. Sansing and Cieglem (1973) described the mass production of spores of several *Aspergillus* and *Penicillium* species on whole loaves of white bread and Podojil *et al.* (1984) quoted the use of millet for the sporulation of *Streptomyces aureofaciens* in the development of inoculum for the chlortetracycline fermentation (see Fig. 6.5).

Sporulation in submerged culture

Many fungi will sporulate in submerged culture provided a suitable medium is employed (Vezina *et al.*, 1965). This technique is more convenient than the use of solid or solidified media because it is easier to operate aseptically and it may be applied on a large scale. The technique was first adopted by Foster *et al.* (1945) who induced submerged sporulation in *Penicillium notatum* by including 2.5% calcium chloride in a defined nitrate–sucrose medium. An example of the use of this technique for the production of inoculum for an industrial fermentation is provided by the griseofulvin process. Rhodes *et al.* (1957) described the conditions necessary for the submerged sporulation of the griseofulvin-producing fungus, *Penicillium patulum*, and

TABLE 6.3. *Solidified media suitable for the sporulation of some representative streptomycetes*

Organism	Product	Medium		References
S. aureofaciens	Tetracycline*	Malt extract (Difco)	1.0%	Williams *et al.* (1974)
		Yeast extract (Difco)	0.4%	
		Glucose	0.4%	
S. erythreus	Erythromycin*	Beef extract (Difco)	0.1%	Williams *et al.* (1974)
		Yeast extract (Difco)	0.1%	
		Casamino acids (Difco)	0.2%	
		Glucose	0.2%	
S. vinaceus	Viomycin*	Corn-steep liquor (50% dry matter)	1.0%	Williams *et al.* (1974)
		Starch	1.0%	
		$(NH_4)_2SO_4$	0.3%	
		NaCl	0.3%	
		$CaCO_3$	0.3%	
S. clavuligerus	Clavulanic* acid	Soluble starch	1.0%	Butterworth (1984)
		K_2HPO_4	0.1%	
		$MgSO_4 \cdot 7H_2O$	0.1%	
		NaCl	0.1%	
		$(NH_4)_2SO_4$	0.2%	
		$CaCO_3$	0.2%	
		$FeSO_4 \cdot 7H_2O$	0.001%	
		$MnCl_2 \cdot 4H_2O$	0.001%	
		$ZnSO_4 \cdot 7H_2O$	0.001%	
S. hygroscopicus	Maridomycin	Soluble starch	1.0%	Miyagawa *et al.* (1979)
		Peptone	0.04%	
		Meat extract	0.02%	
		Yeast extract	0.02%	
		N-Z amine (type A)	0.02%	
		Agar	2.0%	

*Agar content not quoted.

TABLE 6.4. *Solidified media suitable for the sporulation of some representative fungi*

Fungus	Medium	$(g\ dm^{-3})$	References
Penicillium chrysogenum	Glycerol	7.5	Booth (1971)
	Cane molasses	7.5	
	Curbay BG	2.5	
	$MgSO_4 \cdot 7H_2O$	0.05	
	KH_2PO_4	0.06	
	Peptone	5.0	
	NaCl	4.0	
	Agar	20	
Aspergillus niger	Molasses	300	Steel *et al.* (1958)
	KH_2PO_4	0.5	
	Agar	20	

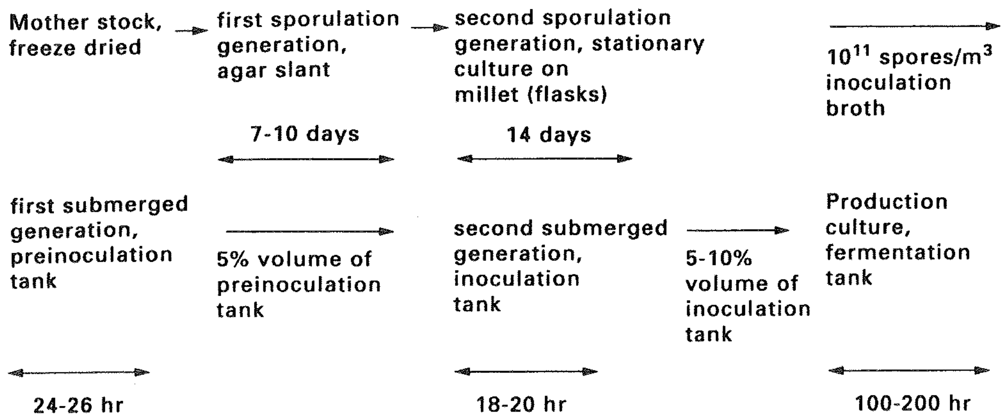

FIG. 6.5. The inoculum development programme for the production of chlortetracycline by *Streptomyces aureofaciens* (Podojil, 1984).

the medium utilized is given in Table 6.5. These authors found that for prolific sporulation the nitrogen level had to be limited to between 0.05 and 0.1% w/v and that good aeration had to be maintained. Also, an interaction was demonstrated between the nitrogen level and aeration in that the lower the degree of aeration the lower the concentration of nitrogen needed to induce sporulation. Submerged sporulation was induced by inoculating 600 cm³ of the above medium, in a 2-dm³ shake flask, with spores from a well-sporulated Czapek-Dox agar culture and incubating at 25° for 7 days. The resulting suspension of spores was then used as a 10% inoculum for a vegetative seed stage in a stirred fermenter, the seed culture subsequently providing a 10% inoculum for the production fermentation. The vegetative seed and production media are given in Table 6.1.

Most actinomycetes do not sporulate in submerged culture (Whitaker, 1992) and, thus, solid or solidified media tend to be used for the production of spore inocula.

The use of the spore inoculum

The stage in an inoculum development programme

TABLE 6.5. *Media for the submerged sporulation of selected fungi*

Rhodes *et al.* (1957): *Penicillium patulum*		
Whey powder, to give	Lactose 3.5% Nitrogen 0.05%	
KH_2PO_4		0.4%
KCl		0.05%
Corn-steep liquor solids to give approx. 0.04% N		0.38%
Foster *et al.* (1945): *Penicillium notatum*		
Sucrose		2.0%
$NaNO_3$		0.6%
KH_2PO_4		0.15%
$MgSO_4 \cdot 7H_2O$		0.05%
$CaCl_2$		2.5%
Vezina *et al.* (1965): *Aspergillus ochraceus*		
Glucose		2.5%
NaCl		2.5%
Corn-steep liquor		0.5%
Molasses		5.0%

at which a large-scale spore inoculum is used varies according to the process; it appears to be common practice that the penultimate stage is so inoculated, but this will depend on the scale of the production fermentation. In the inoculum development programme for the early penicillin fermentation described by Parker (1950) the penultimate stage was inoculated with a spore suspension (from a 'roll-bottle') and this stage may have produced either a vegetative or a submerged spore inoculum for the final fermentation. For the griseofulvin process, Rhodes *et al.* (1957) stated that the spore suspension obtained from the submerged sporulation stage could either be used for direct inoculation of the production fermentation or it could be germinated in an inoculum development medium to yield a vegetative inoculum for the final fermentation. The latter course was preferred and an inoculum volume of 7–10% was used. From Figs 6.4 to 6.6 it can

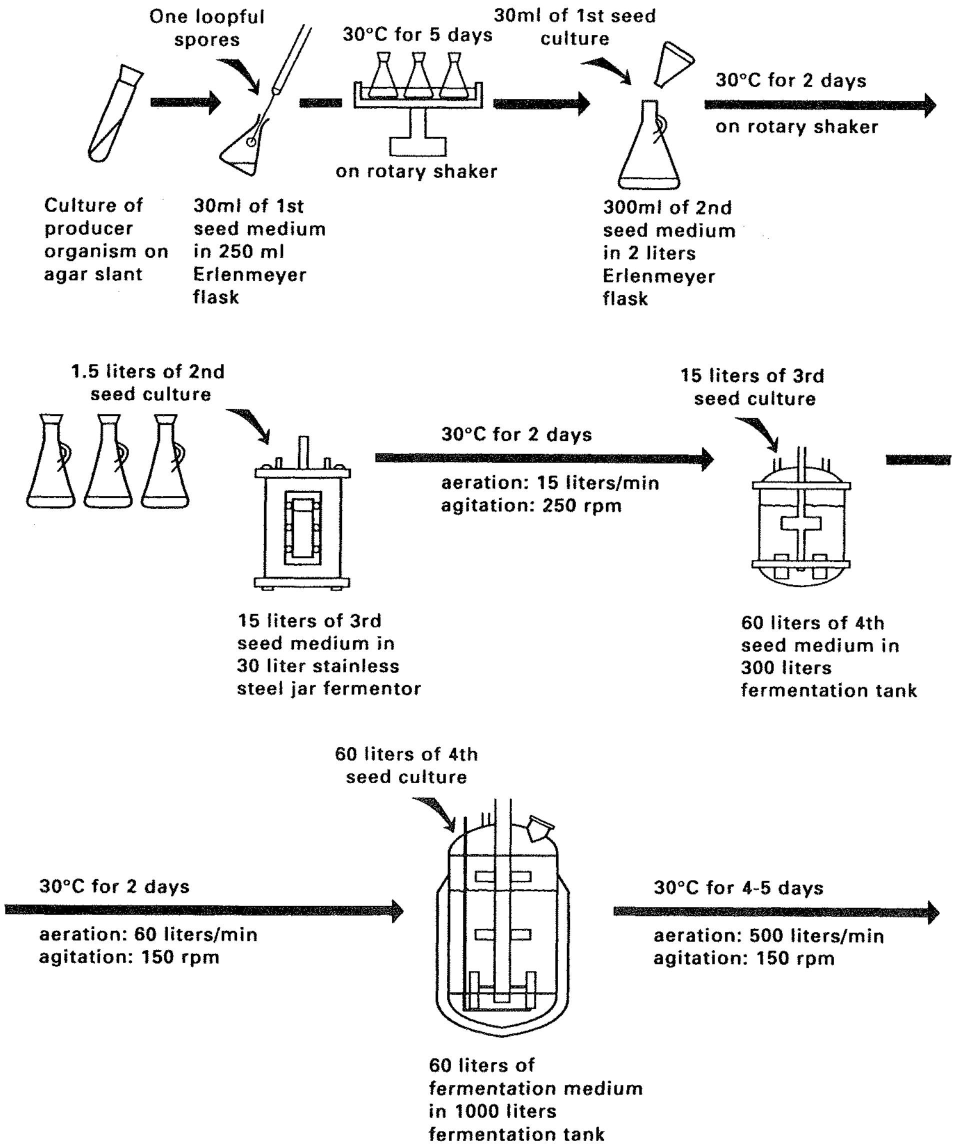

FIG. 6.6. The inoculum development programme for the production of sagamicin by *Micromonospora sagamiensis* (Podojil, 1984).

be seen that in the clavulanic acid process the spore inoculum is used to inoculate the final seed stage, in the chlortetracycline process a vegetative stage is interspersed between the spore inoculated batch and the production fermentation, and in the sagamicin process the spore inoculum is used at a very early stage followed by vegetative growth.

When considering the production of gluconic acid by *Aspergillus niger*, Lockwood (1975) discussed the merits of inoculating the final fermentation directly with a spore suspension as compared with germinating the spores in a seed tank to give a vegetative inoculum. Direct spore inoculation would avoid the cost of installation and operation of the seed tanks whereas the use of germinated spores would reduce the fermentation time of the final stage, thus allowing a greater number of fermentations to be carried out per year. However, labour costs for the production of the vegetative inoculum could be almost as high as for the final fermentation although some of these costs may be recovered, in that gluconic acid produced in the penultimate stage would be recoverable from the final fermentation broth and would contribute to the buffering capacity throughout the fermentation. Thus, Lockwood claimed that the choice of inoculum for the production stage depends on the length of the cycle of the fermentation process, plant size and the availability and cost of labour.

Inoculum development for vegetative fungi

Some fungi will not produce asexual spores and, therefore, an inoculum of vegetative mycelium must be used. *Gibberella fujikuroi* is such a fungus and is used for the commercial production of gibberellin (Borrow *et al.,* 1961). Hansen (1967) described an inoculum development programme for the gibberellin fermentation. Cultures were grown on long slants (25 × 10 mm test tubes) of potato dextrose agar for 1 week at 24°. Growth from three slants was scraped off and transferred to a 9-dm^3 carboy containing 4 dm^3 of a liquid medium composed of 2% glucose, 0.3% $MgSO_4 \cdot 7H_2O$, 0.3% NH_4Cl and 0.3% KH_2PO_4. The medium was aerated for 75 hours at 28° before transfer to a 100-dm^3 seed fermenter containing the same medium.

The major problem in using vegetative mycelium as initial seed is the difficulty of obtaining a uniform, standard inoculum. The procedure may be improved by fragmenting the mycelium in an homogenizer, such as a Waring blender, prior to use as inoculum. This method provides a large number of mycelial particles and therefore a large number of growing points. Worgan (1968) has given a detailed account of the use of this technique in the preparation of inocula for the submerged culture of the higher fungi.

The effect of the inoculum on the morphology of filamentous organisms in submerged culture

When filamentous fungi are grown in submerged culture the type of growth varies from the 'pellet' form, consisting of compact discrete masses of hyphae, to the filamentous form in which the hyphae form a homogeneous suspension dispersed through the medium (Whitaker and Long, 1973). The filamentous type of habit gives rise to an extremely viscous broth which may be very difficult to aerate adequately, whereas the pellet type of habit gives rise to a far less viscous, but also less homogeneous, broth (see Chapter 9). In a pelleted culture there is a danger that the mycelium at the centre of the pellet may be starved of nutrients and oxygen due to diffusion limitations. Also, there is considerable evidence that the morphological form of the organism influences the productivity of the culture, but whether this is due to the phenomena already mentioned or to some form of metabolic control is far from clear. Thus, some fermentations are carried out with the fungus in a filamentous habit, whereas others are carried out with the organism growing as pellets. For example, filamentous growth has been claimed to be optimum for penicillin production by *P. chrysogenum* (Smith and Calam, 1980), whereas pelleted growth has been claimed to be optimum for citric-acid production from *Aspergillus niger* (Al Obaidi and Berry, 1980) and lovastatin from *Aspergillus terreus* (Gbewonyo *et al.,* 1992; see also Chapter 9). The necessity for filamentous growth is taken to the extreme in the ICI–Rank Hovis McDougal mycoprotein process where *Fusarium graminearium* is produced for human consumption. A highly filamentous morphology is required to produce the desired texture in the product which resembles the strength and eating texture of white and soft, red meats (Trinci, 1992). Thus, in this process a median hyphal length of 400 μm is required.

The relevance of this consideration of mycelial morphology to inoculum development is that the morphology may be influenced considerably by both the concentration of spores in a spore inoculum and the inoculum development medium. Usually, a high spore inoculum will tend to produce a dispersed form of growth whilst a low one will favour pellet formation (Foster, 1949). The effect of the concentration of a spore inoculum on the morphology of *P. chrysogenum* is given in Table 6.6. Thus, in the commercial production of fungal products it is critical to grow the organism in the desired morphological form which necessitates the use of an inoculum which achieves this end. If the production fermentation is to be inoculated with a spore suspension then the spore concentration must be such as to produce the production culture in the desired morphological form; if a vegetative inoculum is to be used for the production fermentation then, again, the concentration of its spore inoculum must be such as to produce the vegetative inoculum in the desired morphological form. Although the effects of media on morphological form can be extremely varied dispersed growth is more likely in rich, complex media and pelleted growth tends to occur in chemically defined media (Whitaker and Long, 1973). Thus, the medium used in the spore germination stage must be optimized in terms of the morphology of the inoculum.

An interesting series of experiments on the effects of inoculum conditions on the morphology of *Penicillium citrinum* were reported by Hosobuchi *et al.* (1993). This *Penicillium* species synthesizes compound ML-236B, a precursor of pravastatin which is a cholesterol-lowering drug. Optimum productivity was achieved when the organism grew as compact pellets in the production fermentation. The vegetative inoculum for the production fermentation had to contain an optimum number of short, filamentous propagules in order to initiate pellet formation in the final culture. This was achieved by using a four-stage inoculum development programme (initiated by a spore-inoculated shake flask) with very rich media in the third and fourth cultures. Thus, this system required a dispersed vegetative inoculum to generate a pelleted production fermentation.

The information available on the morphology of actinomycetes in submerged culture is very limited compared with that on fungi. However, Whitaker (1992) has reviewed the area and it is obvious that actinomycetes are capable of producing a wide range of morphological types. Also, it appears to be accepted that a dispersed mycelial morphology is desirable for most industrial actinomycete fermentations. Mycelial forms have been shown to be desirable for the production of streptomycin by *Streptomyces griseus* and turimycin by *S. hygroscopicus*, whereas the pelleted form of *S. nigrificans* was better for glucose isomerase production (Whitaker, 1992). As already discussed for the fungi, the concentration of spores in the inoculum has also been shown to influence the morphology of certain streptomycetes (Lawton *et al.,* 1989). These workers also demonstrated that medium composition and the shear forces operating during culture also affect morphological form. Thus, the principles applied to the optimization of fungal inoculum development regimes are also relevant to actinomycete processes. Hunt and Stieber (1986) described the optimization of the inoculum regime of a small-scale streptomycete cephamycin C fermentation. Pellet formation was observed to be detrimental to product formation and

TABLE 6.6. *The effect of spore concentration and medium on the morphology and penicillin productivity of* Penicillium chrysogenum

Medium	Spore concentration in the medium	Morphology
Camici *et al.* (1952):		
Corn-steep dextrin	More than 10×10^5 dm^{-3}	Filamentous
	Less than 10×10^5 dm^{-3}	Pellets
Czapek-dox	More than 3.0×10^5 dm^{-3}	Filamentous
	Less than 3.0×10^5 dm^{-3}	Pellets
Glucose, lactose and ammonium lactate	More than 2.0×10^5 dm^{-3}	Filamentous
	Less than 2.0×10^5 dm^{-3}	Pellets
Spore concentration in the inoculum (cm^{-3})	**Penicillin yield (units cm^{-3})**	**Morphology**
Calam (1976):		
10^2	500	Dense pellets
10^3	1800	Dense pellets
2×10^3	4000	Open pellets
10^4	5000	Filamentous

the key factor in establishing the correct form appeared to be the concentration of iron in the seed medium, a higher iron concentration giving the optimum inoculum.

THE ASEPTIC INOCULATION OF PLANT FERMENTERS

The inoculation of plant-scale fermenters may involve the transfer of culture from a laboratory fermenter, or spore suspension vessel, to a plant fermenter, or the transfer from one plant fermenter to another. Obviously, it is extremely important that the fermentation is not contaminated during inoculation but if the process is a contained one then it is equally important that the process organism does not escape. Thus, the nature of the inoculation system will be dictated by the containment category of the process (see Chapter 7).

At Containment levels 1 and B2, the addition of inoculum must be carried out in such a way that release of micro-organisms is restricted. This should be done by aseptic piercing of membranes or connections with steam locks. At Containment level 2 and B3/4, no micro-organisms must be released during inoculation or other additions. In order to meet these stringent requirements, all connections must be screwed or clamped and all pipelines must be steam sterilizable (Werner, 1992).

Inoculation from a laboratory fermenter or a spore suspension vessel

Several systems have been described in the literature which are suitable for inoculating fermentations requiring only Good Industrial Large Scale Practice (GILSP, see Chapter 7). To prevent contamination during the transfer process it is essential that both vessels be maintained under a positive pressure and the inoculation port be equipped with a steam supply. Meyrath and Suchanek (1972) described a system for the inoculation of a plant fermenter from a laboratory vessel. The apparatus is shown in Fig. 6.7. The connecting point A is normally covered with the blank plug a and prior to inoculation this plug is slightly loosened, valve E closed and valve F opened to allow steam to exit at A. Valve F is then closed and E opened so that when a is removed sterile air will be released from the vessel. After removal, plug a is placed in strong disinfectant. Blank plug b is then removed and a coupling made between B and A. Valve E is closed and an air line attached to point C, establishing a pressure inside the inoculum fermenter greater than in the plant vessel. Valve E is then opened and the inoculum will be forced into the plant fermenter. After closing valve E the inoculum fermenter may be removed, plug a replaced, and the line steamed out by opening valve F. This system may be modified by using the quick connection devices which are now available (see Chapter 7).

Jackson (1958) has described a very similar system for the introduction of a spore suspension into a plant fermenter. The apparatus is shown in Fig. 6.8 and its operation is identical to that described by Meyrath and Suchanek.

Parker (1958) described a more complex system for inoculation of a plant fermenter from a spore-suspension vessel. The apparatus is shown in Fig. 6.9. The sterile spore-suspension vessel is batched with the spore suspension in the sterile room. The plant vessel, containing the medium and with blank plugs screwed on at A and B, is sterilized by steam injection. The plugs A and B are slightly loosened to allow steam to emit for 20 minutes when the whole system is under steam pressure. The blanks are then tightened, the valves E and G shut and sterile air is allowed into the plant fermenter by opening valves D, F and C. The spore suspension vessel is loosely connected at A and B and valves D, H, I and C closed and E, F and G opened. After 20 minutes steaming A and B are tightened and G and E are closed. D is then opened to establish a positive pressure in the pipework. When the pipework has cooled, the pressure in the plant vessel is reduced to approximately 5 psi, valve F is closed and H, I and C are opened. This procedure allows the spore suspension to be forced into the fermenter. Valves D, H, I

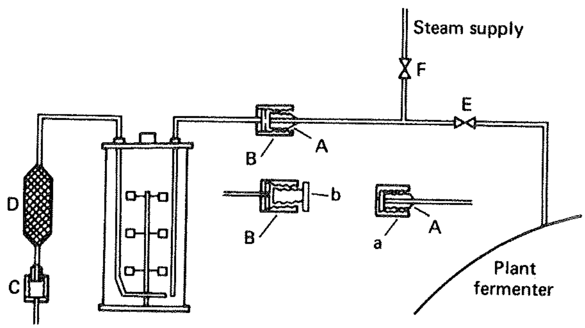

Fig. 6.7. Inoculation of a plant fermenter from a laboratory fermenter (Meyrath and Suchanek, 1972).

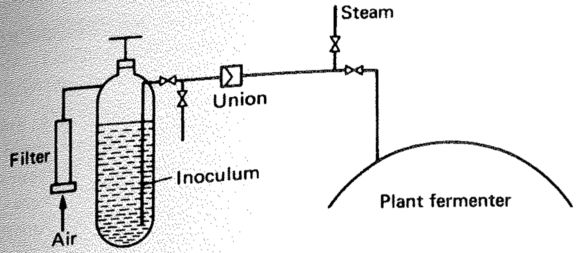

FIG. 6.8. Inoculation of a plant fermenter from a spore suspension vessel (Jackson, 1958).

and C are closed and the suspension vessel replaced by blank plugs at A and B and the pipework steamed by opening valves E and G.

GILSP and category 1 fermentations may also be inoculated by aseptic piercing of a membrane. In this system the inoculum vessel is connected to an inoculating needle assembly (as shown in Fig. 6.10) and the sterile needle pierces a rubber septum set into a fermenter port. However, the use of a needle presents safety problems and many companies prohibit such systems. Also, aerosols may be created on the removal of a needle.

The inoculation of level 2 or B3/4 contained fermentations requires the use of modified systems. It must be possible to steam sterilize all the pipework

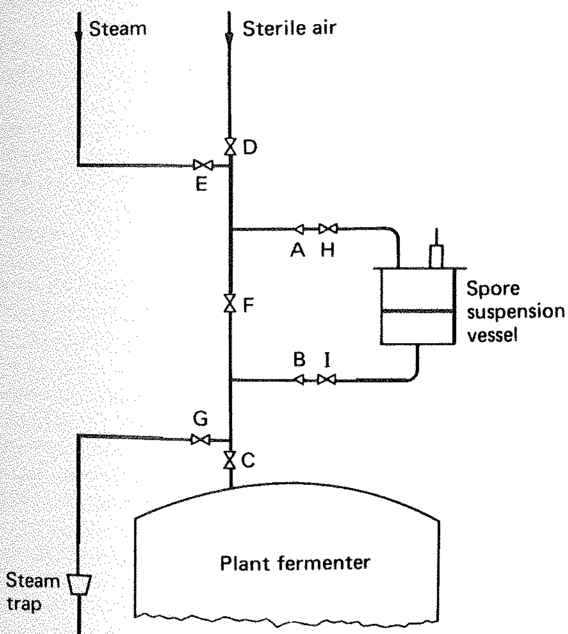

FIG. 6.9. The inoculation of a plant fermenter from a spore suspension vessel (Parker, 1958).

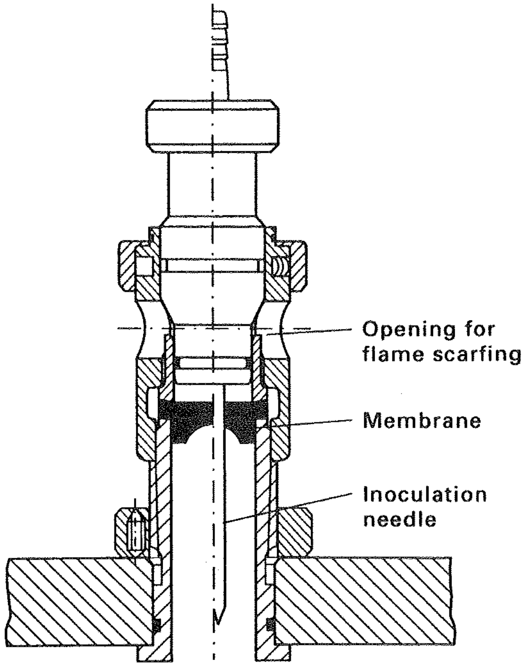

FIG. 6.10. Needle inoculation device (Werner, 1992).

after the inoculation and the condensate from the sterilization must be collected in a kill tank (see Chapters 5 and 7). The inoculation flask is then removed after inoculation and sterilized in an autoclave. One such system is shown in Fig. 6.11 (Vranch, 1990).

Inoculation from a plant fermenter

Figure 6.12 illustrates the system described by Parker (1958). The two vessels are connected by a flexible pipeline A–B. The batched fermenter is sterilized by steam injection via valves G and J, valves D, I, A, B, H, E and F being open and valve C closed. Valves H and I lead to steam traps for the removal of condensate. After 20 minutes at the desired pressure the steam supply is switched off at J and G and the steam-trap valves I and H are closed. F, E and D are left open so that the connecting pipeline fills with sterile medium. The medium in the fermenter is sparged with sterile air and when it has cooled to the desired temperature the pressure in the seed tank is increased to at least 10 psi whilst the pressure in the fermenter is reduced to about 2 psi. Valve C is opened and the inoculum is forced into the production vessel. After inoculation is

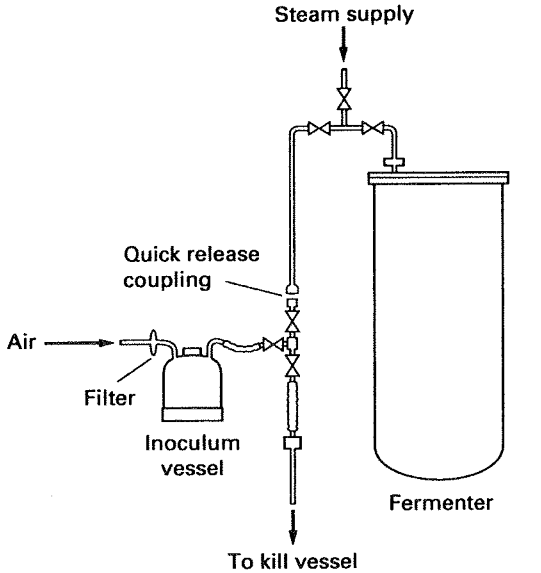

FIG. 6.11. Inoculum system suitable for contained fermentations (Vranch, 1990).

complete the connecting pipeline is resterilized before it is removed. For a contained fermentation the condensate from the two steam traps attached to valves I and H would be directed to a kill tank.

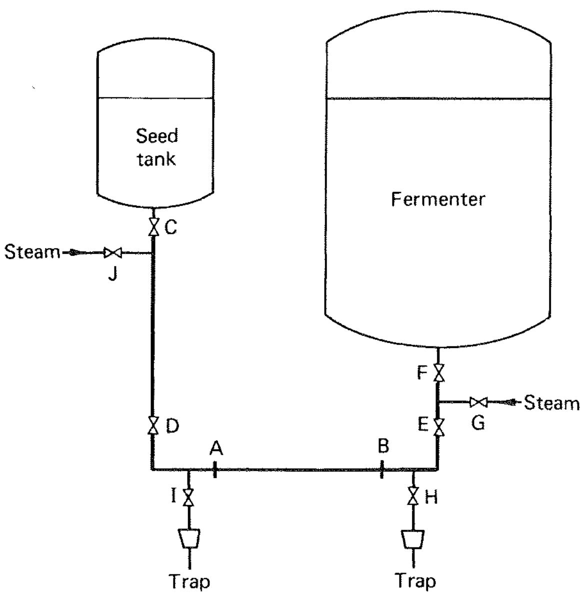

FIG. 6.12. Inoculation of a plant fermenter from another plant fermenter (Parker, 1958).

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CHAPTER 7

Design of a Fermenter

INTRODUCTION

A RESEARCH team led by Chaim Weizmann in Great Britain during the First World War (1914–1918) developed a process for the production of acetone by a deep liquid fermentation using *Clostridium acetobutylicum* which led to the eventual use of the first truly large-scale aseptic fermentation vessels (Hastings, 1978). Contamination, particularly with bacteriophages, was often a serious problem, especially during the early part of a large-scale production stage. Initially, no suitable vessels were available and attempts with alcohol fermenters fitted with lids were not satisfactory as steam sterilization could not be achieved at atmospheric pressure. Large mild-steel cylindrical vessels with hemispherical tops and bottoms were constructed that could be sterilized with steam under pressure. Since the problems of aseptic additions of media or inocula had been recognized, steps were taken to design and construct piping, joints and valves in which sterile conditions could be achieved and maintained when required. Although the smaller seed vessels were stirred mechanically, the large production vessels were not, and the large volumes of gas produced during the fermentation continually agitated the vessel contents. Thus, considerable expertise was built up in the construction and operation of this aseptic anaerobic process for production of acetone–butanol.

The first true large-scale aerobic fermenters were used in Central Europe in the 1930s for the production of compressed yeast (de Becze and Liebmann, 1944). The fermenters consisted of large cylindrical tanks with air introduced at the base via networks of perforated pipes. In later modifications, mechanical impellers were used to increase the rate of mixing and to break up and disperse the air bubbles. This procedure led to the compressed-air requirements being reduced by a factor of 5. Baffles on the walls of the vessels prevented a vortex forming in the liquid. Even at this time it was recognized that the cost of energy necessary to compress air could be 10 to 20% of the total production cost. As early as 1932, Strauch and Schmidt patented a system in which the aeration tubes were provided with water and steam for cleaning and sterilizing.

Prior to 1940, the other important fermentation products besides bakers' yeast were ethanol, glycerol, acetic acid, citric acid, other organic acids, enzymes and sorbose (Johnson, 1971). These processes used highly selective environments such as acidic or anaerobic conditions or the use of an unusual substrate, resulting in contamination being a relatively minor problem compared with the acetone fermentation or the subsequent aerobic antibiotic fermentations.

The decision to use submerged culture techniques for penicillin production, where aseptic conditions, good aeration and agitation were essential, was a very important factor in forcing the development of carefully designed and purpose-built fermentation vessels. In 1943, when the British government decided that surface culture production was inadequate, none of the fermentation plants were immediately suitable for deep fermentation, although the Distillers Company solvent plant at Bromborough only needed aeration equipment to make it suitable for penicillin production (Hastings, 1971). Construction work on the first large-scale plant to produce penicillin by deep fermentation was started on 15th September 1943, at Terre Haute in the United States of America, building steel fermenters with working volumes of 54,000 dm^3 (Callahan, 1944). The plant was operational on 30th January 1944. Unfortunately, no other construction details were quoted for the fermenters.

Initial agitation studies in baffled stirred tanks to identify variables were also reported at this time by

Cooper *et al.* (1944), Foust *et al.* (1944) and Miller and Rushton (1944). Cooper's work had a major influence on the design of later fermenters.

BASIC FUNCTIONS OF A FERMENTER FOR MICROBIAL OR ANIMAL CELL CULTURE

The main function of a fermenter is to provide a controlled environment for the growth of micro-organisms or animal cells, to obtain a desired product. In designing and constructing a fermenter a number of points must be considered:

1. The vessel should be capable of being operated aseptically for a number of days and should be reliable in long-term operation and meet the requirements of containment regulations.
2. Adequate aeration and agitation should be provided to meet the metabolic requirements of the micro-organism. However, the mixing should not cause damage to the organism.
3. Power consumption should be as low as possible.
4. A system of temperature control should be provided.
5. A system of pH control should be provided.
6. Sampling facilities should be provided.
7. Evaporation losses from the fermenter should not be excessive.
8. The vessel should be designed to require the minimal use of labour in operation, harvesting, cleaning and maintenance.
9. Ideally the vessel should be suitable for a range of processes, but this may be restricted because of containment regulations.
10. The vessel should be constructed to ensure smooth internal surfaces, using welds instead of flange joints whenever possible.
11. The vessel should be of similar geometry to both smaller and larger vessels in the pilot plant or plant to facilitate scale-up.
12. The cheapest materials which enable satisfactory results to be achieved should be used.
13. There should be adequate service provisions for individual plants (see Table 7.1).

The first two points are probably the most critical. It is obvious from the above points that the design of a fermenter will involve co-operation between experts in microbiology, biochemistry, chemical engineering, mechanical engineering and costing. Although many different types of fermenter have been described in the literature, very few have proved to be satisfactory for industrial aerobic fermentations. The most commonly used ones are based on a stirred upright cylinder with sparger aeration. This type of vessel can be produced in a range of sizes from one dm^3 to thousands of dm^3. Figures 7.1 and 7.2 are diagrams of typical mechanically agitated and aerated fermenters with one and three multi-bladed impellers respectively. Tables 7.2 and 7.3 give geometrical ratios of various of the dimensions which have been quoted in the literature for a variety of sizes of vessel.

TABLE 7.1. *Service provisions for a fermentation plant*

Compressed air
Sterile compressed air (at 1.5 to 3.0 atmospheres)
Chilled water (12 to 15°)
Cold water (4°)
Hot water
Steam (high pressure)
Steam condensate
Electricity
Stand-by generator
Drainage of effluents
Motors
Storage facilities for media components
Control and monitoring equipment for fermenters
Maintenance facilities
Extraction and recovery equipment
Accessibility for delivery of materials
Appropriate containment facilities

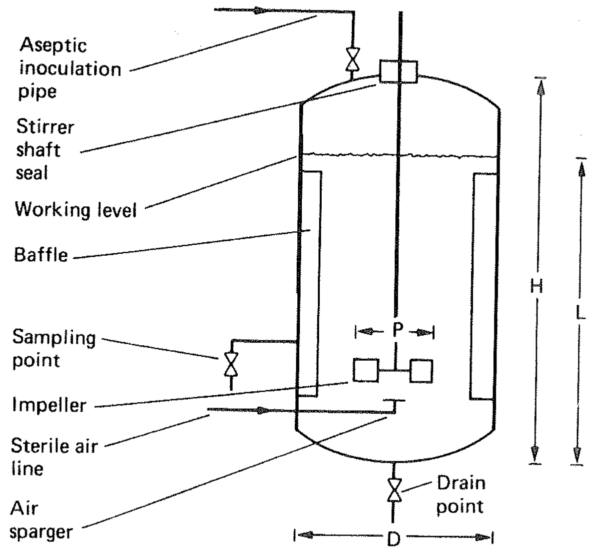

FIG. 7.1. Diagram of a fermenter with one multi-bladed impeller.

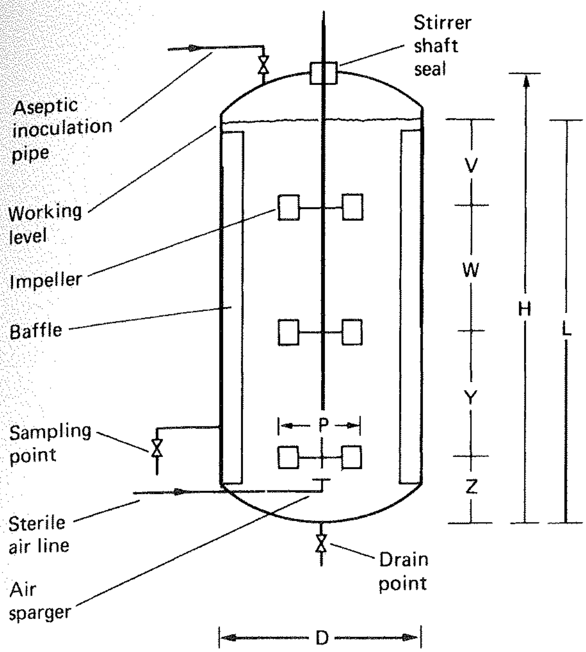

FIG. 7.2. Diagram of a fermenter with three multi-bladed impellers.

At this stage the discussion will be concerned with stirred, aerated vessels for microbial cell culture. More varied shapes are commonly used for alcohol, biomass production, animal cell culture and effluent treatment and will be dealt with later in this chapter.

ASEPTIC OPERATION AND CONTAINMENT

Aseptic operation involves protection against contamination and it is a well established and understood concept in the fermentation industries, whereas containment involves prevention of escape of viable cells from a fermenter or downstream equipment and is much more recent in origin. Containment guidelines were initiated during the 1970s (East *et al.*, 1984; Flickinger and Sansone, 1984).

To establish the appropriate degree of containment which will be necessary to grow a micro-organism, it, and in fact the entire process, must be carefully assessed for potential hazards that could occur should there be accidental release. Different assessment procedures are used depending on whether or not the organism contains foreign DNA (genetically engineered). Once the hazards are assessed, an organism can be classified into a hazard group for which there is an appropriate level of containment. The procedure which has been adopted within the European Community is outlined in Fig. 7.3. Non-genetically engineered organisms may be placed into a hazard group (1 to 4) using criteria to assess risk such as those given by Collins (1992):

1. The known pathogenicity of the micro-organism.
2. The virulence or level of pathogenicity of the micro-organism — are the diseases it causes mild or serious?
3. The number of organisms required to initiate an infection.
4. The routes of infection.
5. The known incidence of infection in the community and the existence locally of vectors and potential reserves.
6. The amounts or volumes of organisms used in the fermentation process.
7. The techniques or processes used.
8. Ease of prophylaxis and treatment.

A detailed discussion of assessment of risk and ways to reduce it is given by Winkler and Park (1992).

TABLE 7.2. *Details of geometrical ratios of fermenters with single multi-blade impellers (see Fig. 7.1)*

Dimension	Steel and Maxon (1961)	Wegrich and Shurter (1963)	Blakeborough (1967)
Operating volume	250 dm^3	12 dm^3	—
Liquid height (L)	55 cm	27 cm	—
L/D (tank diameter)	0.72	1.1	1.0–1.5
Impeller diameter (P/D)	0.4	0.5	0.33
Baffle width/D	0.10	0.08	0.08–0.10
Impeller height/D	—	—	0.33

TABLE 7.3. *Details of geometrical ratios of fermenters with three multi-bladed impellers (see Fig. 7.2)*

Dimension	Jackson (1958)	Aiba *et al.* (1973)	Paca *et al.* (1976)
Operating volume	—	100,000 dm^3 (total)	170 dm^3
Liquid height (L)	—	—	150 cm
L/D (tank diameter)	—	—	1.7
Impeller diameter	0.34–0.5	0.4	0.33
Baffle width/D	0.08–0.1	0.095	0.098
Impeller height/D	0.5	0.24	0.37
P/V	0.5–1.0	—	0.74
P/W	0.5–1.0	0.85	0.77
P/Y	0.5–1.0	0.85	0.77
P/Z	—	2.1	0.91
H/D	1.0–1.6	2.2	2.95

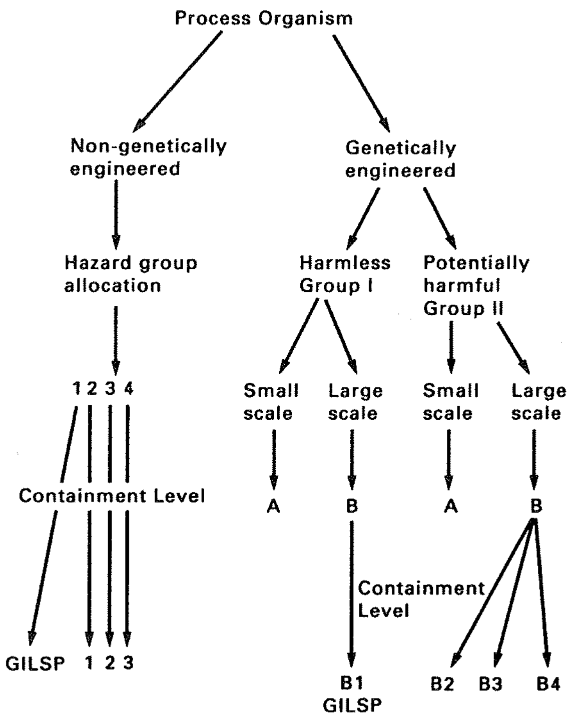

FIG. 7.3. Categorization of a process micro-organism and designation of its appropriate level of containment at research or industrial sites within the European Federation of Biotechnology (GILSP = Good Industrial Large Scale Practice).

Once the organism has been allocated to a hazard group, the appropriate containment requirements can be applied (see Table 7.4). Hazard group 1 organisms used on a large scale only require Good Industrial Large Scale Practice (GILSP). Processes in this category need to be operated aseptically but no containment steps are necessary, including prevention of escape of organisms. If the organism is placed in Hazard group 4 the stringent requirements of level 3 will have to be met before the process can be operated. Details of hazard categories for a range of organisms can be obtained from Frommer *et al.* (1989).

Genetically engineered organisms are classified as either harmless (Group I) or potentially harmful (Group II). The process is then classified as either small scale (A: less than 10 dm^3) or large scale (B: more than 10 dm^3) according to guidelines which can be found in the Health and Safety Executive document (1993). Therefore large scale processes fall into two categories, IB or IIB. IB processes require containment level B1 and are subject to GILSP, whereas IIB processes are further assessed to determine the most suitable containment level, ranging from B2 to B4 as outlined in Table 7.4. Levels B2 to B4 correspond to levels 1 to 3 for non-genetically engineered organisms.

In future it is possible, under new legislation, that no distinction will be made between non-genetically engineered and genetically engineered organisms. The key factor will be whether the organism is harmless or potentially harmful, regardless of its genetic constitution. Containment would then be decided using the scheme which is currently being used for genetically engineered organisms.

Other hazard-assessment systems for classifying organisms have been introduced in many other countries. Production and research workers must abide by appropriate local official hazard lists. Problems can occur when different official bodies place the same organism in different hazard categories. In 1989, the European

TABLE 7.4. *Summary of safety precautions for biotechnological operations in the European Federation for Biotechnology (EFB)* (Frommer *et al.*, 1989)

Procedures	GILSP*	Containment category 1	Containment category 2	Containment category 3
Written instructions and code of practice	+	+	+	+
Biosafety manual	–	+	+	+
Good occupational hygiene	+	+	+	+
Good Microbiological Techniques (GMT)	–	+	+	+
Biohazard sign	–	+	+	+
Restricted access	–	+	+	+
Accident reporting	+	+	+	+
Medical surveillance	–	+	+	+
Primary containment: Operation and equipment				
Work with viable micro-organisms should take place in closed systems (CS), which minimize (m) or prevent (p) the release of cultivated micro-organisms	–	m	p	p
Treatment of exhaust air or gas from CS	–	m	p	p
Sampling from CS	–	m	p	p
Addition of materials to CS, transfer of cultivated cells	–	m	p	p
Removal of material, products and effluents from CS	–	m	p	p
Penetration of CS by agitator shaft and measuring devices	–	m	p	p
Foam-out control	–	m	p	p
Secondary containment: Facilities				
Protective clothing appropriate to the risk category	+	+	+	+
Changing/washing facility	+	+	+	+
Disinfection facility	–	+	+	+
Emergency shower facility	–	–	+	+
Airlock and compulsory shower facilities	–	–	–	+
Effluents decontaminated	–	–	+	+
Controlled negative pressure	–	–	–	+
HEPA filters in air ducts	–	–	+	+
Tank for spilled fluids	–	–	+	+
Area hermetically sealable	–	–	–	+

*Unless required for product quality, –, not required; +, required.
m, minimize release. The level of contamination of air, working surface and personnel shall not exceed the level found during microbiological work applying Good Microbiological Techniques.
p, prevent release. No detectable contamination during work should be found in the air, working surfaces and personnel.

Federation for Biotechnology were aware of this problem with non-recombinant micro-organisms and produced a consensus list (Frommer *et al.*, 1989).

Most micro-organisms used in industrial processes are in the lowest hazard group which only require GILSP, although some organisms used in bacterial and viral vaccine production and other processes are categorized in higher groups. There is an obvious incentive for industry to use an organism which poses a low risk as this minimizes regulatory restrictions and reduces the need for expensive equipment and associated containment facilities (Schofield, 1992). The costs of some containment design features are discussed in Chapter 12.

In this chapter, where appropriate, a distinction will be made between equipment which is used to maintain aseptic conditions for GILSP and those modifications which are needed to meet physical containment requirements in order to grow specific cultures which have been classified in higher hazard groups. Details on design for containment have been given by East *et al.* (1984), Flickinger and Sansone (1984), Walker *et al.* (1987), Turner (1989), Hambleton *et al.* (1991), Leaver and Hambleton (1992), Vranch (1992) and Werner (1992).

Overall containment categorization

In this chapter the emphasis will be on containment levels which might be obtained from particular designs of fermenters and associated equipment. However this is only one aspect of assessment. To meet the standards of the specific level of containment it will also be necessary to consider the procedures to be used, staff training, the facilities in the laboratory and factory, downstream processing, effluent treatment, work practice, maintenance, etc. It will be necessary to ensure that **all** these aspects are of a sufficiently high standard to meet the levels of containment deemed necessary for a particular process by a government regulatory body. If these are met, then the process can be operated.

BODY CONSTRUCTION

Construction materials

In fermentations with strict aseptic requirements it is important to select materials that can withstand repeated steam sterilization cycles. On a small scale (1 to 30 dm^3) it is possible to use glass and/or stainless steel. Glass is useful because it gives smooth surfaces, is non-toxic, corrosion proof and it is usually easy to examine the interior of the vessel. Two basic types of fermenter are used:

1. A glass vessel with a round or flat bottom and a top flanged carrying plate (Fig. 7.4). The large glass containers originally used were borosilicate battery jars (Brown and Peterson, 1950). All vessels of this type have to be sterilized by autoclaving. Cowan and Thomas (1988) state that the largest practical diameter for glass fermenters is 60 cm.
2. A glass cylinder with stainless-steel top and bottom plates (Fig. 7.5). These fermenters may be sterilized *in situ*, but 30 cm diameter is the upper size limit to safely withstand working pressures (Solomons, 1969). Vessels with two stainless steel plates cost approximately 50% more than those with just a top plate.

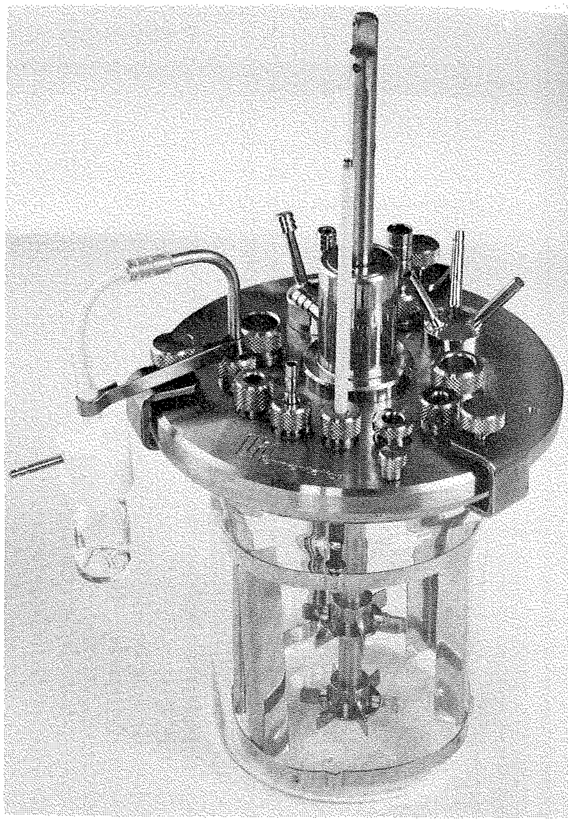

FIG. 7.4. Glass fermenter with a top-flanged carrying plate (Inceltech L.H. Reading, England).

At pilot and large scale (Figs 7.6 and 7.7), when all fermenters are sterilized *in situ*, any materials used will have to be assessed on their ability to withstand pressure sterilization and corrosion and on their potential toxicity and cost. Walker and Holdsworth (1958), Solomons (1969) and Cowan and Thomas (1988) have discussed the suitability of various materials used in the construction of fermenters. Pilot-scale and industrial-

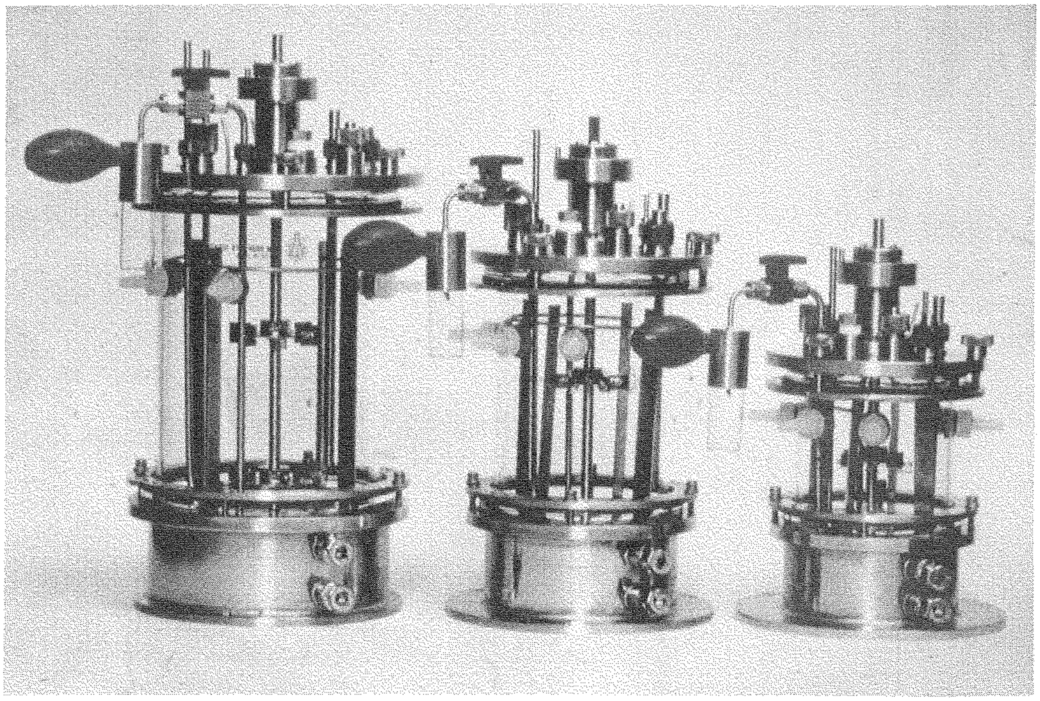

FIG. 7.5. Three glass fermenters with top and bottom plates (New Brunswick Scientific, Hatfield, England).

scale vessels are normally constructed of stainless steel or at least have a stainless-steel cladding to limit corrosion. The American Iron and Steel Institute (AISI) states that steels containing less than 4% chromium are classified as steel alloys and those containing more than 4% are classified as stainless steels. Mild steel coated with glass or phenolic epoxy materials has occasionally been used (Gordon *et al.*, 1947; Fortune *et al.*, 1950; Buelow and Johnson, 1952; Irving, 1968). Wood, plastic and concrete have been used when contamination was not a problem in a process (Steel and Miller, 1970).

Walker and Holdsworth (1958) stated that the extent of vessel corrosion varied considerably and did not appear to be entirely predictable. Athough stainless steel is often quoted as the only satisfactory material, it has been reported that mild-steel vessels were very satisfactory after 12-years use for penicillin fermentations (Walker and Holdsworth, 1958) and mild steel clad with stainless steel has been used for at least 25 years for acetone–butanol production (Spivey, 1978). Pitting to a depth of 7 mm was found in a mild-steel fermenter after 7-years use for streptomycin production (Walker and Holdsworth, 1958).

The corrosion resistance of stainless steel is thought to depend on the existence of a thin hydrous oxide film on the surface of the metal. The composition of this film varies with different steel alloys and different manufacturing process treatments such as rolling, pickling or heat treatment. The film is stabilized by chromium and is considered to be continuous, non-porous, insoluble and self healing. If damaged, the film will repair itself when exposed to air or an oxidizing agent (Cubberly *et al.*, 1980).

The minimum amount of chromium needed to resist corrosion will depend on the corroding agent in a particular environment, such as acids, alkalis, gases, soil, salt or fresh water. Increasing the chromium content enhances resistance to corrosion, but only grades of steel containing at least 10 to 13% chromium develop an effective film. The inclusion of nickel in high percent chromium steels enhances their resistance and improves their engineering properties. The presence of molybdenum improves the resistance of stainless steels to solutions of halogen salts and pitting by chloride ions in brine or sea water. Corrosion resistance can also be improved by tungsten, silicone and other elements (Cubberly *et al.*, 1980; Duurkoop, 1992).

AISI grade 316 steels which contain 18% chromium, 10% nickel and 2–2.5% molybdenum are now com-

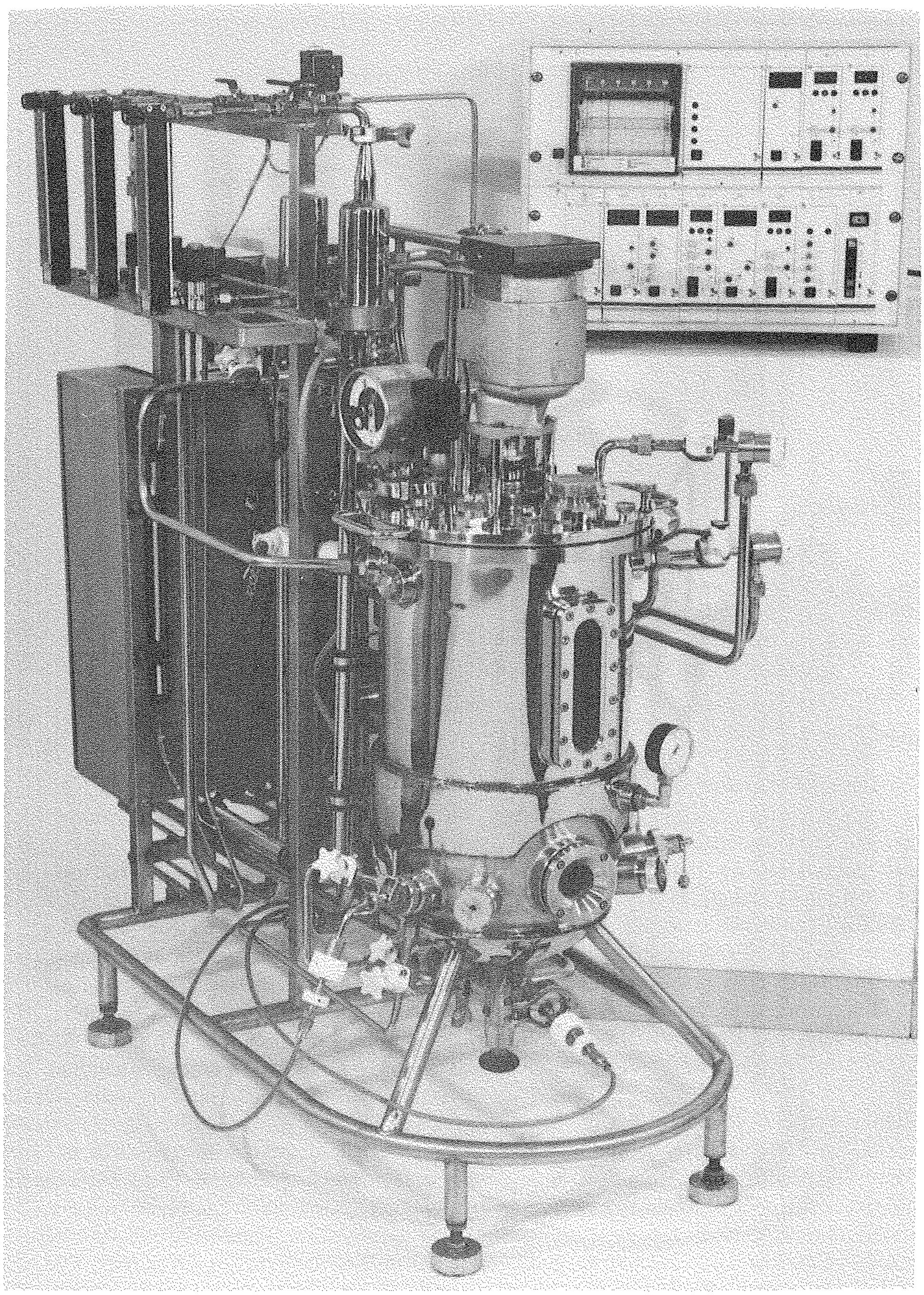

FIG. 7.6. Stainless steel fully automatic 10 dm^3 fermenter sterilizable-*in-situ* (LSL, Luton, UK).

monly used in fermenter construction. Table 7.5 (Durkoop, 1992) gives the classification systems used to identify stainless steels of AISI 316 grades in the U.S.A. and similar procedures used in Germany, France, the United Kingdom, Japan and Sweden. In a citric acid fermentation where the pH may be 1 to 2 it will be necessary to use a stainless steel with 3–4% molybdenum (AISI grade 317) to prevent leaching of heavy

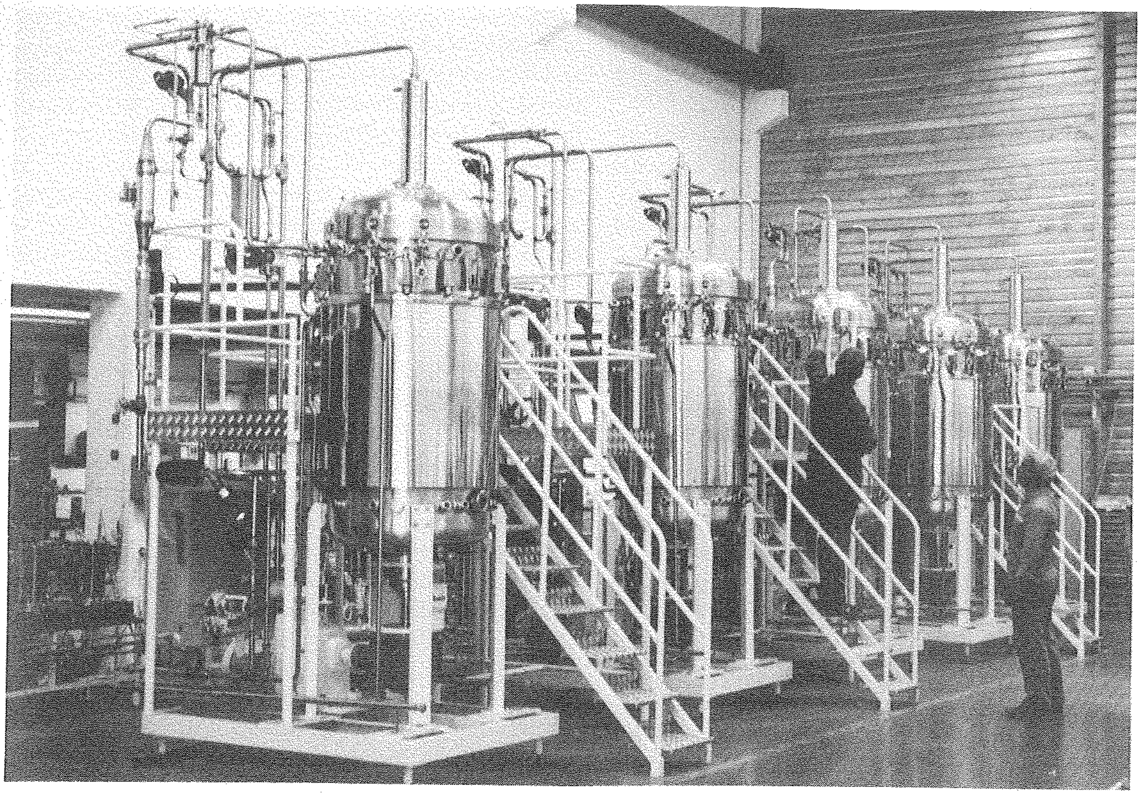

FIG. 7.7. Stainless steel pilot plant fermenters (LSL, Luton, UK).

TABLE 7.5 *Classification systems used for stainless steels of 316 grades* (Duurkoop, 1992)

Germany	DIN	France	Great Britain	Japan	Sweden	U.S.A.	
Werkstoff Nummer		AFNOR	B.S.	JIS	SS	AISI	UNS
1.4401	X2 CrNiMo 17 12 2	Z 6 CND 17.11	316 S 16 316 S 31	SUS 316	2347	316	S31600
1.4404	X2 CrNiMo 17 13 2 G-X CrNiMo 18 10	Z 2 CND 18.13 Z 2 CND 17 12 Z 2 CND 19.10 M	316 S 11 316 S 12	SUS 316L	2348	316L	S31603
1.4406	X2 CrNiMo 17 12 2	Z 2 CND 17.12 Az	316 S 61	SUS 316LN	—	316LN	S31653
1.4435	X2 CrNiMo 18 14 3	Z 2 CND 17.13	316 S 11 316 S 12	SCS 16 SUS 316L	2353	316L	S31603

metals from the steel which would interfere with the fermentation. AISI Grade 304, which contains 18.5% chromium and 10% nickel, is now used extensively for brewing equipment.

The thickness of the construction material will increase with scale. At 300,000 to 400,000 dm^3 capacity, 7-mm plate may be used for the side of a vessel and 10 mm plate for the top and bottom, which should be hemispherical to withstand pressures.

At this stage it is important to consider the ways in which a reliable aseptic seal is made between glass and glass, glass and metal or metal and metal joints such as between a fermenter vessel and a detachable top or base plate. With glass and metal, a seal can be made with a compressible gasket, a lip seal or an 'O' ring (Fig. 7.8). With metal to metal joints only an 'O' ring is suitable. This is placed in a groove, machined in either the end plate, the fermenter body or both. This seal ensures that a good liquid- and/or gas-tight joint is maintained in spite of the glass or metal expanding or contracting at different rates with changes in temperature during a sterilization cycle or an incubation cycle. Nitryl or butyl rubbers are normally used for these seals as they will withstand fermentation process conditions. The properties of different rubbers for seals are discussed by Buchter (1979) and Martini (1984). These rubber seals have a finite life and should be checked regularly for damage or perishing. Before purchasing a fermenter it is important to check that standard sized 'O' rings can be purchased cheaply from local suppliers.

A single 'O' ring seal is adequate for GILSP and levels 1 and B2, a double 'O' ring seal is required for levels 2 and B3 and a double 'O' ring seal with steam between the seals (steam tracing) is necessary for levels 3 and B4 (Chapman, 1989; Hambleton *et al.*, 1991). In the U.S.A., however, simple seals are used at containment levels comparable with the U.K. B3 (Titchener-Hooker *et al.*, 1993).

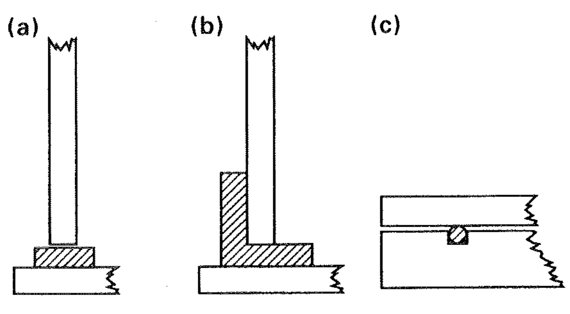

FIG. 7.8. Joint seals for glass–glass, glass–metal and metal–metal; (a) gasket; (b) lip seal; (c) 'O' ring in groove.

Titchener-Hooker *et al.* (1993) criticized Chapman's proposal to use two seals without a steam trace for a number of reasons including:

a. Double seals are more difficult to assemble correctly.
b. It is difficult to detect failure of one seal of a pair during operation or assembly.
c. Neither of the two seals can be tested independently.
d. Dead spaces between two seals must be considered to be contaminated.

Leaver (1994) and Titchener-Hooker *et al.* (1993) consider that correctly fitted single static seals can provide adequate containment for most processes and double static seals with a steam trace should be strictly limited to the small number of processes for which an extreme level of protection may be required.

Temperature control

Normally in the design and construction of a fermenter there must be adequate provision for temperature control which will affect the design of the vessel body. Heat will be produced by microbial activity and mechanical agitation and if the heat generated by these two processes is not ideal for the particular manufacturing process then heat may have to be added to, or removed from, the system. On a laboratory scale little heat is normally generated and extra heat has to be provided by placing the fermenter in a thermostatically controlled bath, or by the use of internal heating coils or a heating jacket through which water is circulated or by a silicone heating jacket. The silicone jacket consists of a double silicone rubber mat with heating wires between the two mats; it is wrapped around the vessel and held in place by Velcro strips (Applikon, 1989).

Once a certain size has been exceeded, the surface area covered by the jacket becomes too small to remove the heat produced by the fermentation. When this situation occurs internal coils must be used and cold water is circulated to achieve the correct temperature (Jackson, 1990). Different types of fermentation will influence the maximum size of vessel that can be used with jackets alone.

It is impossible to specify accurately the necessary cooling surface of a fermenter since the temperature of the cooling water, the sterilization process, the cultivation temperature, the type of micro-organism and the energy supplied by stirring can vary considerably in different processes. A cooling area of 50 to 70 m^2 may be taken as average for a 55,000 dm^3 fermenter and with a coolant temperature of 14° the fermenter may be cooled from 120° to 30° in 2.5 to 4 hours without stirring. The consumption of cooling water in this size of vessel during a bacterial fermentation ranges from 500 to 2000 $dm^3\ h^{-1}$, while fungi might need 2000 to 10,000 $dm^3\ h^{-1}$ (Muller and Kieslich, 1966), due to the lower optimum temperature for growth.

To make an accurate estimate of heating/cooling requirements for a specific process it is important to consider the contributing factors. An overal energy balance for a fermenter during normal operation can be written as:

$$Q_{met} + Q_{ag} + Q_{gas} = Q_{acc} + Q_{exch} + Q_{evap} + Q_{sen} \tag{7.1}$$

where Q_{met} = heat generation rate due to microbial metabolism,
Q_{ag} = heat generation rate due to mechanical agitation,
Q_{gas} = heat generation rate due to aeration power imput,
Q_{acc} = heat accumulation rate by the system,
Q_{exch} = heat transfer rate to the surroundings and/or heat exchanger,
Q_{evap} = heat loss rate by evaporation,
Q_{sen} = rate of sensible enthalpy gain by the flow streams (exit − inlet).

This equation can be rearranged as:

$$Q_{exch} = Q_{met} + Q_{ag} + Q_{gas} - Q_{acc} - Q_{sen} - Q_{evap} \tag{7.2}$$

Q_{exch} is the heat which will have to be removed by a cooling system.

Atkinson and Mavituna (1991b) have presented data to estimate Q_{met} for a range of substrates, methods to calculate power input for Q_{ag} and Q_{gas}, a formula to calculate the sensible heat loss for flow streams (Q_{sen}) and a method to calculate the heat loss due to evaporation (Q_{evap}). Cooney *et al.* (1969) determined representative low and high heats of fermentation values for *Bacillus subtilis* grown on molasses at 37° (Table 7. 6). They concluded that Q_{evap} and Q_{sen} are small contributory factors and Q_{acc} = 0 in a steady-state system. Q_{evap} can also be eliminated by using a saturated air stream at the temperature of the broth.

TABLE 7.6. *Representative low and high values of calculated heats (kcal dm^{-3} h^{-1}) of fermentation for* Bacillus subtilis *on molasses at* 37° (Coooney *et al.*, 1969)

	Heats of fermentation					
	Q_{acc}	Q_{ag}	Q_{evap}	Q_{sen}	Q_{exch}	Q_{met}
Low	3.81	3.32	0.023	0.005	0.61	1.12
High	11.3	3.31	0.045	0.010	0.65	8.65

When designing a large fermenter, the operating temperature and flow conditions will determine Q_{evap} and Q_{sen}, the choice of agitator, its speed and the aeration rate will determine Q_{ag} and the sparger design and aeration rate will determine Q_{gas}.

The cooling requirements (jacket and/or pipes) to remove the excess heat from a fermenter may be determined by the following formula:

$$Q_{exch} = U \cdot A \cdot \Delta T \tag{7.3}$$

where A = the heat transfer surface available, m^2,
Q = the heat transferred, W,
U = the overall heat transfer coefficient, W/m^2K,
ΔT = the temperature difference between the heating or cooling agent and the mass itself, K.

The coefficient U represents the conductivity of the system and it depends on the vessel geometry, fluid properties, flow velocity, wall material and thickness (Scragg, 1991). $1/U$ is the overall resistance to heat transfer (analogous to 1/K for gas–liquid transfer; Chapter 9). It is the reciprocal of the overall heat-transfer coefficient. It is defined as the sum of the individual resistances to heat transfer as heat passes from one fluid to another and can be expressed as:

$$\frac{1}{U} = \frac{1}{h_o} + \frac{1}{h_i} + \frac{1}{h_{of}} + \frac{1}{h_{if}} + \frac{1}{h_w} \tag{7.4}$$

where h_o = outside film coefficient, W/m^2K,
h_i = inside film coefficient, W/m^2K,
h_{of} = outside fouling film coefficient, W/m^2K,
h_{if} = inside fouling film coefficient, W/m^2K,
h_w = wall heat transfer coefficient $= k/x$, W/m^2K,
k = thermal conductivity of the wall, W/mK,
x = wall thickness, m.

There is more detailed discussion of U in Bailey and Ollis (1986), Atkinson and Mavituna (1991b) and Scragg (1991).

Atkinson and Mavituna (1991b) have given three methods to determine ΔT (the temperature driving force) depending on the operating circumstances. If one side of the wall is at a constant temperature, as is often the case in a stirred fermenter, and the coolant temperature rises in the direction of the coolant flow along a cooling coil, an arithmetic mean is appropriate:

$$\Delta T_{am} = \frac{(T_f - T_e) + (T_f - T_i)}{2} \quad (7.5)$$

$$= \frac{T_f - (T_e - T_i)}{2} \quad (7.6)$$

where T_f = the bulk liquid temperature in the vessel,
T_e = the temperature of the coolant entering the system,
T_i = the temperature of the coolant leaving the system.

If the fluids are in counter- or co-current flow and the temperature varies in both fluids then a log mean temperature difference is appropriate:

$$\Delta T_m = \Delta T_e - \Delta T_i / \ln (\Delta T_e / \Delta T_i) \quad (7.7)$$

where ΔT_e = the temperature of the coolant entering,
ΔT_i = the temperature of the coolant leaving.

If the flow pattern is more complex than either of the two previous situations then the log mean temperature difference defined in equation (7.7) is multiplied by an appropriate dimensionless factor which has been evaluated for a number of heat-exchanger systems by McAdams (1954).

Appropriate techniques have just been discussed to obtain values for Q_{exch}, U and ΔT (or ΔT_{am} or ΔT_m). If equation (7.3) is now rearranged:

$$A = \frac{Q_{exch}}{U \Delta T}. \quad (7.8)$$

Substituting in this equation makes it possible to calculate the heat-transfer surface necessary to obtain adequate temperature control.

AERATION AND AGITATION

Aeration and agitation will be considered in detail in Chapter 9. In this chapter it should be stated that the primary purpose of aeration is to provide micro-organisms in submerged culture with sufficient oxygen for metabolic requirements, while agitation should ensure that a uniform suspension of microbial cells is achieved in a homogeneous nutrient medium. The type of aeration–agitation system used in a particular fermenter depends on the characteristics of the fermentation process under consideration. Although fine bubble aerators without mechanical agitation have the advantage of lower equipment and power costs, agitation may be dispensed with only when aeration provides sufficient agitation, i.e. in processes where broths of low viscosity and low total solids are used (Arnold and Steel, 1958). Thus, mechanical agitation is usually required in fungal and actinomycete fermentations. Non-agitated fermentations are normally carried out in vessels of a height/diameter ratio of 5:1. In such vessels aeration is sufficient to produce high turbulence, but a tall column of liquid does require greater energy input in the production of the compressed air (Muller and Kieslich, 1966; Solomons, 1980).

The structural components of the fermenter involved in aeration and agitation are:

(a) The agitator (impeller).
(b) Stirrer glands and bearings.
(c) Baffles.
(d) The aeration system (sparger).

The agitator (impeller)

The agitator is required to achieve a number of mixing objectives, e.g. bulk fluid and gas-phase mixing, air dispersion, oxygen transfer, heat transfer, suspension of solid particles and maintaining a uniform environment throughout the vessel contents. It should be possible to design a fermenter to achieve these conditions; this will require knowledge of the most appropriate agitator, air sparger, baffles, the best positions for nutrient feeds, acid or alkali for pH control and antifoam addition. There will also be a need to specify agitator size and number, speed and power imput (see also Chapter 9).

Agitators may be classified as disc turbines, vaned discs, open turbines of variable pitch and propellers, and are illustrated in Fig. 7.9. The disc turbine consists of a disc with a series of rectangular vanes set in a vertical plane around the circumference and the vaned disc has a series of rectangular vanes attached vertically to the underside. Air from the sparger hits the underside of the disc and is displaced towards the vanes where the air bubbles are broken up into smaller bubbles. The vanes of a variable pitch open turbine and

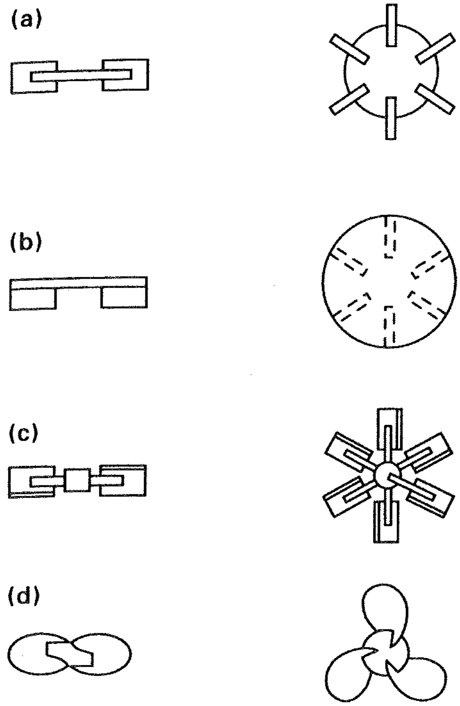

FIG. 7.9. Types of agitators: (a) disc turbine; (b) vaned disc; (c) open turbine, variable pitch; (d) marine propeller (Solomons, 1969).

the blades of a marine propeller are attached directly to a boss on the agitator shaft. In this case the air bubbles do not initially hit any surface before dispersion by the vanes or blades.

Four other modern agitator developments, the Scaba 6SRGT, the Prochem Maxflo T, the Lightning A315 and the Ekato Intermig (Figs 7.10 and 7.11), which are derived from open turbines, will also be discussed for energy conservation and use in high-viscosity broths.

Since the 1940s a Rushton disc turbine of one-third the fermenter diameter has been considered the optimum design for use in many fermentation processes. It had been established experimentally that the disc turbine was most suitable in a fermenter since it could break up a fast air stream without itself becoming flooded in air bubbles (Finn, 1954). This flooding condition is indicated when the bulk flow pattern in the vessel normally associated with the agitator design (radial with the Rushton turbine) is lost and replaced by a centrally flowing air–broth plume up the middle of the vessel with a liquid flow as an annulus (Nienow *et al.*, 1985; see also Chapter 9). The propeller and the open turbine flood when V_s (superficial velocity, i.e. volumetric air flow rate/cross-sectional area of fermenter) exceeds 21 m h^{-1}, whereas the flat blade turbine can tolerate a V_s of up to 120 m h^{-1} before being flooded, when two sets are used on the same shaft. Besides being flooded at a lower V_s than the disc turbine, the propeller is also less efficient in breaking up a stream of air bubbles and the flow it produces is axial rather than radial (Cooper *et al.*, 1944). The disc turbine was thought to be essential for forcing the sparged air into the agitator tip zone where bubble break up would occur.

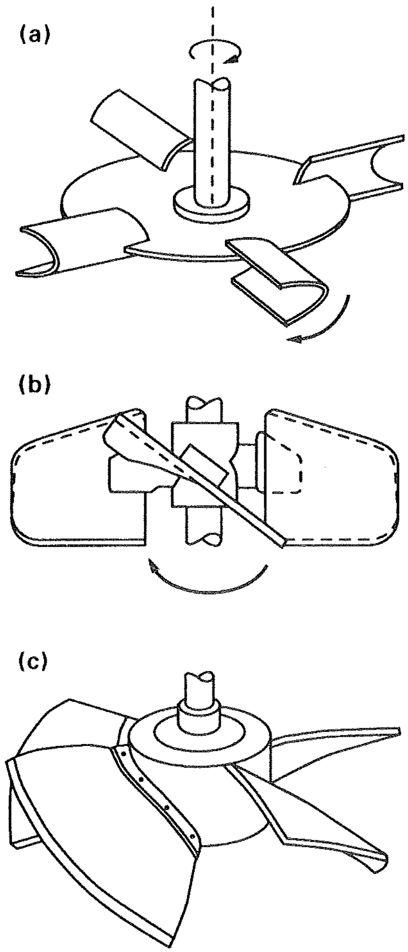

FIG. 7.10. Diagrams of (a) Scaba agitator; (b) Lightnin' A315 agitator (four blades) and (c) Prochem Maxflo T agitator (four, five or six blades) (Nienow, 1990).

In other studies it has been shown that bubble break up occurs in the trailing vortices associated with all agitator types which give rise to gas-filled cavities and

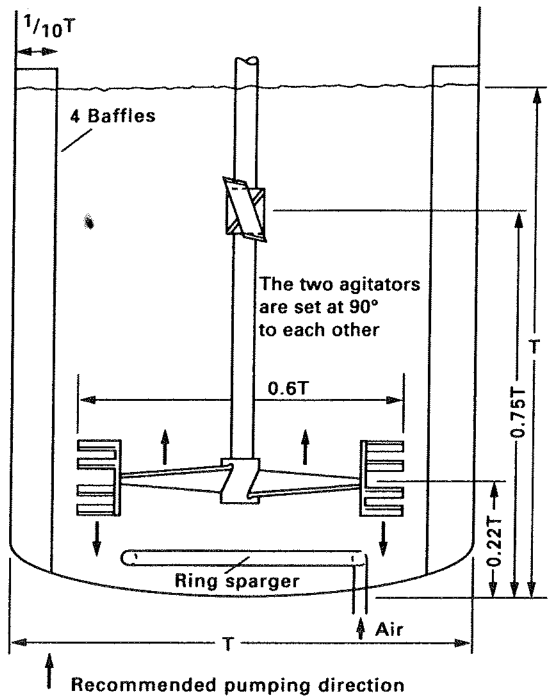

FIG. 7.11. Arrangement for a pair of Intermig agitators. Relative dimensions are given as a proportion of the fermenter vessel diameter (T) (Nienow, 1990).

provided the agitator speed is high enough, good gas dispersion will occur in low-viscosity broths (Smith, 1985). It has been also shown that similar oxygen-transfer efficiencies are obtained at the same power input per unit volume, regardless of the agitator type.

In high-viscosity broths, gas dispersion also occurs from gas filled cavities trailing behind the rotating blades, but this is not sufficient to ensure satisfactory bulk blending of all the vessel contents. When the cavities are of maximum size, the impeller appears to be rotating in a pocket of gas from which little actual dispersion occurs into the rest of the vessel (Nienow and Ulbrecht, 1985).

Recently, a number of agitators have been developed to overcome problems associated with efficient bulk blending in high-viscosity fermentations (Figs 7.10 and 7.11). The Scaba 6SRGT agitator is one which, at a given power input, can handle a high air flow rate before flooding (Nienow, 1992). This radial-flow agitator is also better for bulk blending than a Rushton turbine, but does not give good top to bottom blending in a large fermenter which leads to lower concentrations of oxygen in broth away from the agitators and higher concentrations of nutrients, acid or alkali or antifoam near to the feed points.

Another is the Prochem Maxflo agitator. It consists of four, five or six hydrofoil blades set at a critical angle on a central hollow hub (Gbewonyo *et al.*, 1986; Buckland *et al.*, 1988). A high hydrodynamic thrust is created during rotation, increasing the downwards pumping capacity of the blades. This design minimizes the drag forces associated with rotation of the agitator such that the energy losses due to drag are low. This leads to a low power number. The recommended agitator to vessel diameter ratio is greater than 0.4. When the agitator was used with a 800-dm^3 *Streptomyces* fermentation, the maximum power requirement at the most viscous stage was about 66% of that with Rushton turbines. The fall in power was also less in a 14,000-dm^3 fermentation. The oxygen-transfer efficiency was also significantly improved. It was thought that an improvement in bulk mixing was another contributing factor.

Intermig agitators (Fig. 7.11) made by Ekato of Germany are more complex in design. Two units are used (with agitator diameter to vessel diameter ratios of 0.6 to 0.7) instead of a single Rushton turbine because their power number is so low (Nienow, 1992). A large-diameter air sparger is used to optimize air dispersion (see later section on spargers). The loss in power is less than when aerating with a Rushton turbine. Air dispersion starts from the air cavities which form on the wing tips of the agitator blades. In spite of the downwards pumping direction of the wings, the cavities extend horizontally from the back of the agitator blades, reducing the effectiveness of top to bottom mixing in a vessel.

Cooke *et al.* (1988) and Nienow (1990, 1992) give comparisons of performance of Rushton turbines with some of the newer designs.

These new turbine designs make it possible to replace Rushton turbines by larger low power agitators which do not lose as much power when aerated, are able to handle higher air volumes without flooding and give better bulk blending and heat transfer characteristics in more viscous media (Nienow, 1990). However, there can be mechanical problems which are mostly of a vibrational nature.

Good mixing and aeration in high viscosity broths may also be achieved by a dual impeller combination, where the lower impeller acts as the gas disperser and the upper impeller acts primarily as a device for aiding circulation of vessel contents. This has been discussed by Nienow and Ulbrecht (1985) and in Chapter 9.

Steel and Maxon (1966) tested a multi-rod mixing impeller. In a 15,000-dm^3 vessel, the same novobiocin yield and oxygen availability rate were obtained at about half of the power required by a standard turbine-stirred fermenter, but this type of impeller does not appear to have come into general use.

Stirrer glands and bearings

The satisfactory sealing of the stirrer shaft assembly has been one of the most difficult problems to overcome in the construction of fermentation equipment which can be operated aseptically for long periods. A number of different designs have been developed to obtain aseptic seals. The stirrer shaft can enter the vessel from the top, side (Richards, 1968) or bottom of the vessel. Top entry is most commonly used, but bottom entry may be advantageous if more space is needed on the top plate for entry ports, and the shorter shaft permits higher stirrer speeds to be used by eliminating the problem of the shaft whipping at high speeds. Originally, bottom entry stirrers were considered undesirable as the bearings would be submerged. Chain *et al.* (1952) successfully operated vessels of this type, and they have since been used by many other workers. Mechanical seals can be used for bottom entry provided that they are routinely maintained and replaced at recommended intervals (Leaver and Hambleton, 1992).

One of the earliest stirrer seals described was that used by Rivett, Johnson and Peterson (1950) in a laboratory fermenter (Fig. 7.12). A porous bronze bearing for a 13-mm shaft was fitted in the centre of the fermenter top and another in a yoke directly above it. The bearings were pressed into steel housings, which screwed into position in the yoke and the fermenter top. The lower bearing and housing were covered with a skirt-like shield having a 6.5 mm overhang which rotated with the shaft and prevented air-borne contaminants from settling on the bearing and working their way through it into the fermenter.

Four basic types of seal assembly have been used: the stuffing box (packed-gland seal), the simple bush seal, the mechanical seal and the magnetic drive. Most modern fermenter stirrer mechanisms now incorporate mechanical seals instead of stuffing boxes and packed glands. Mechanical seals are more expensive, but are more durable and less likely to be an entry point for contaminants or a leakage point for organisms or products which should be contained. Magnetic drives,

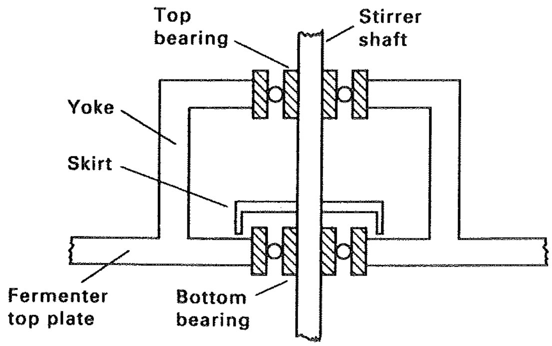

FIG. 7.12. A simple stirrer seal based on a description given by Rivett *et al.* (1950).

which are also quite expensive, are being used in some animal cell culture vessels.

THE STUFFING BOX (PACKED-GLAND SEAL)

The stuffing box (Fig. 7.13) has been described by Chain *et al.* (1954). The shaft is sealed by several layers of packing rings of asbestos or cotton yarn, pressed against the shaft by a gland follower. At high stirrer speeds the packing wears quickly and excessive pressure may be needed to ensure tightness of fit. The packing may be difficult to sterilize properly because of unsatisfactory heat penetration and it is necessary to check and replace the packing rings regularly. Parker (1950) described a split stuffing box with a lantern ring. Steam under pressure was continually fed into it. Chain *et al.* (1954) used two stuffing boxes on the agitator shaft with a space in between kept filled with steam. Although, at one time, stuffing box-bearings were commonly used in large-scale vessels, operational problems, particularly contamination, have led to their replacement by mechanical seal bearings for many processes. However, these seals are sufficient for the requirements of GILSP containment (Werner, 1992).

THE MECHANICAL SEAL

The mechanical seal assembly (Figs 7.14 and 7.15) is now commonly used in both small and large fermenters. The seal is composed of two parts, one part is stationary in the bearing housing, the other rotates on the shaft, and the two components are pressed together by springs or expanding bellows. The two meeting surfaces have to be precision machined, the moving surface normally consists of a carbon-faced unit while the stationary unit is of stellite-faced stainless steel. Steam condensate is used to lubricate and cool the seals during operation and serve as a containment

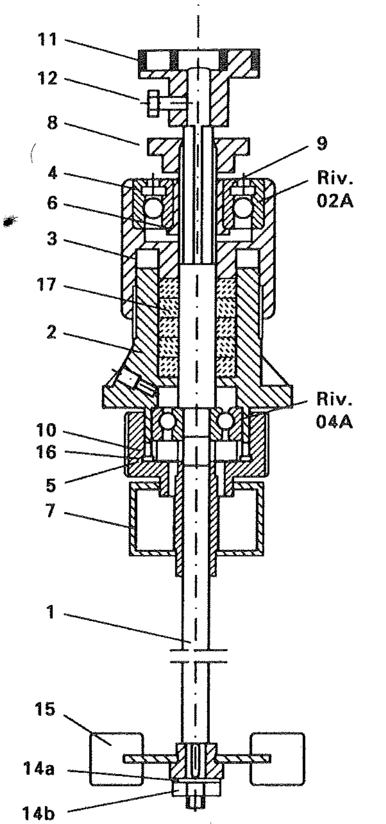

FIG. 7.13. Packed-gland stirrer seal (Chain *et al.*, 1954) (Components: 1, agitator shaft; 2, stuffing box; 3, upper cap; 4, lock ring; 5, lower cap; 6, chuck; 7, greasecup; 8, lock ring; 9, lock nut; 10, distance ring; 11, half coupling; 12, half coupling; 14a, washer; 14b, nut; 15, impeller; 16, shim; 17, packing rings).

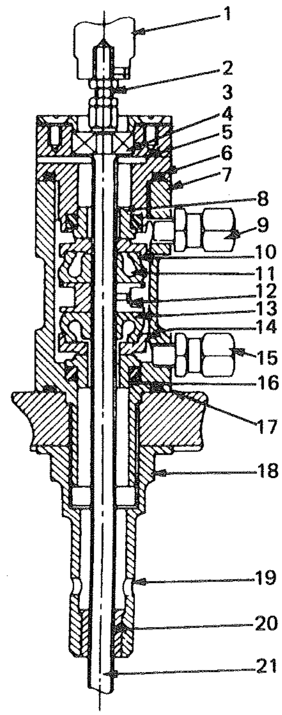

FIG. 7.14. Mechanical seal assembly (Elsworth *et al.*, 1958). (Components; 1, flexible coupling; 2, stirrer shaft; 3, bearing housing; 4, ball journal fit on mating parts; 5, two slots for gland leaks, only one shown; 6, 'O'-ring seal; 7, seal body; 8, stationary counter-face sealed to body with square-section gasket; 9, exit port for condensate, fitted with unequal stud coupling; 10, rotating counter-face; 11, bellows; 12, shaft muff; 13, as 11; 14, as 10; 15, entry port for condensate, as 9; 16, as 8; 17, as 6; 18, shaft bush support; 19, leak holes; 20, Ferobestos bush; 21, ground shaft).

barrier. Single mechanical seals are used with a steam barrier in fermenters for primary containment at level 1 or B2 (Werner, 1992), whereas double mechanical seals are typically used in vessels with the outer seal as a backup for the inner seal for primary containment at level 2 or B3 (Werner, 1992; Leaver and Hambleton, 1992; Fig. 7.15). At level 2 or B3 the condensate is piped to a kill tank. Monitoring of the steam condensate flowing out of the seal is an effective way for checking for seal failure. Disinfectants are alternatives for flushing the seals (Werner, 1992).

MAGNETIC DRIVES

The problems of providing a satisfactory seal when the impeller shaft passes through the top or bottom plate of the fermenter may be solved by the use of a magnetic drive in which the impeller shaft does not pierce the vessel (Cameron and Godfrey, 1969). A magnetic drive (Fig. 7.16) consists of two magnets: one driving and one driven. The driving magnet is held in bearings in a housing on the outside of the head plate and connected to a drive shaft. The internal driven magnet is placed on one end of the impeller shaft and held in bearings in a suitable housing on the inner surface of the headplate. When multiple ceramic magnets have been used, it has been possible to transmit power across a gap of 16 mm. Using this drive, water can be stirred in baffled vessels of up to 300-dm^3 capacity at speeds of 300 to 2000 rpm. It would be necessary to establish if adequate power could be transmitted between magnets to stir viscous mould broths or when wanting high oxygen transfer rates in bacterial cultures. Walker *et al.* (1987) have described the development of a magnetic drive suitable for microbial fermentations up to 1500 dm^3 which could be used when

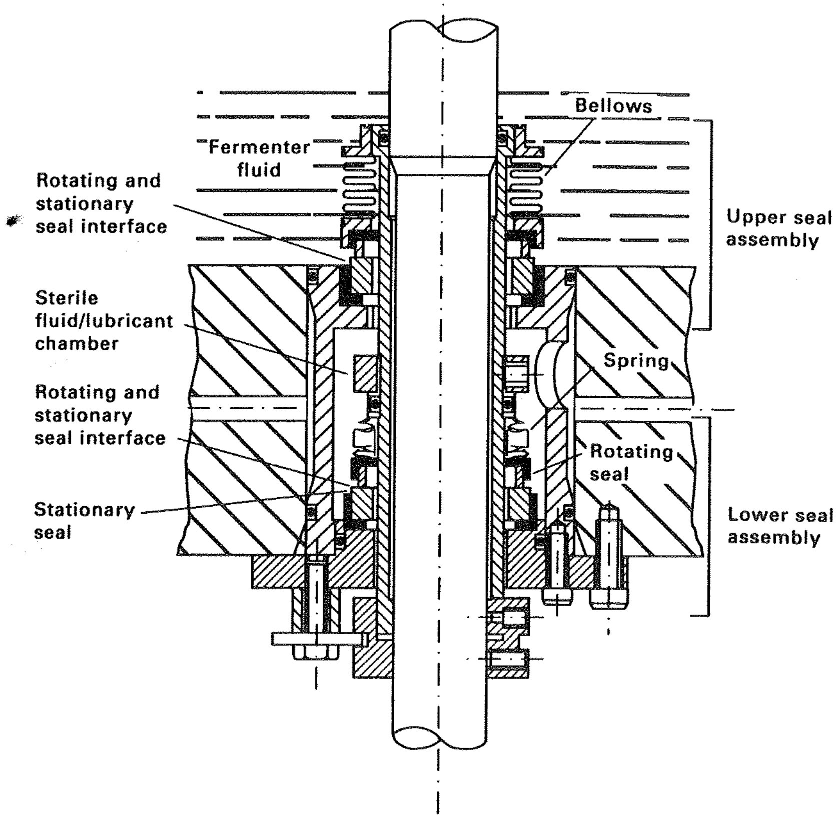

FIG. 7.15. Double mechanical seal (New Brunswick Scientific, Hatfield, England).

higher containment levels are specified. The stirring mechanism is ideal for animal cell culture to minimize the chances of potential contamination. This application is discussed later in this chapter.

Baffles

Four baffles are normally incorporated into agitated vessels of all sizes to prevent a vortex and to improve aeration efficiency. In vessels over 3-dm^3 diameter six or eight baffles may be used (Scragg, 1991). Baffles are metal strips roughly one-tenth of the vessel diameter and attached radially to the wall (see Figs 7.1 and 7.2 and Tables 7.2 and 7.3). The agitation effect is only slightly increased with wider baffles, but drops sharply with narrower baffles (Winkler, 1990). Walker and Holdsworth (1958) recommended that baffles should be installed so that a gap existed between them and the vessel wall, so that there was a scouring action around and behind the baffles thus minimizing microbial growth on the baffles and the fermenter walls. Extra cooling coils may be attached to baffles to improve the cooling capacity of a fermenter without unduly affecting the geometry.

The aeration system (sparger)

A sparger may be defined as a device for introducing air into the liquid in a fermenter. Three basic types of sparger have been used and may be described as the porous sparger, the orifice sparger (a perforated pipe) and the nozzle sparger (an open or partially closed pipe). A combined sparger–agitator may be used in laboratory fermenters (Fig. 7.17) and is discussed briefly in a later section.

POROUS SPARGER

The porous sparger of sintered glass, ceramics or metal, has been used primarily on a laboratory scale in

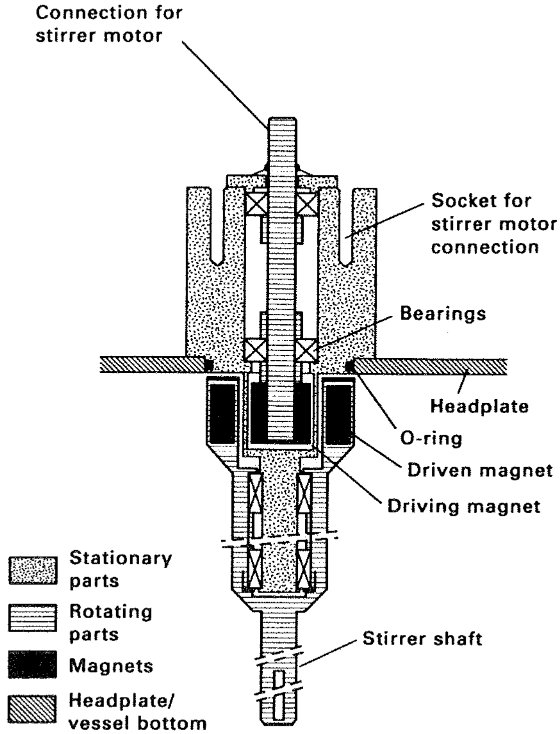

FIG. 7.16. Diagram of magnetically coupled top stirrer assembly (Applikon Dependable Instruments, Schiedam, Netherlands).

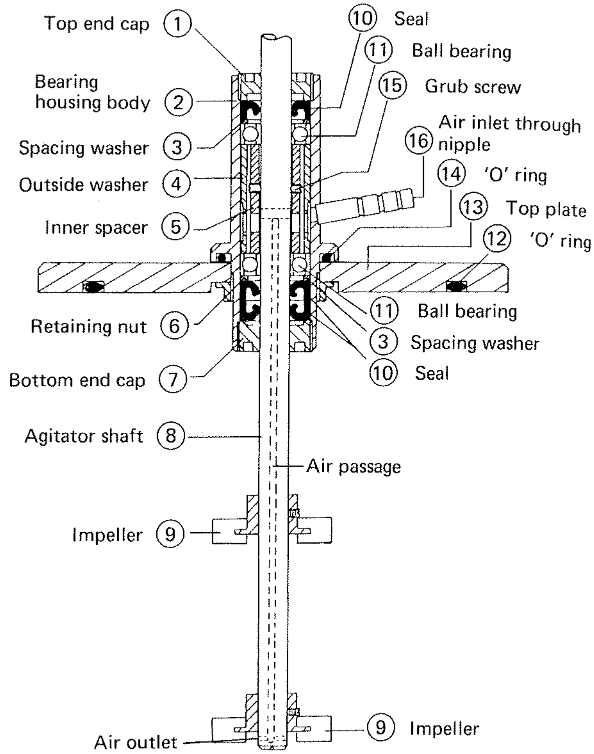

FIG. 7.17. Diagram of bearing housing with combined agitator–sparger (Inceltech L.H. Reading, England).

non-agitated vessels. The bubble size produced from such spargers is always 10 to 100 times larger than the pore size of the aerator block (Finn, 1954). The throughput of air is low because of the pressure drop across the sparger and there is also the problem of the fine holes becoming blocked by growth of the microbial culture.

ORIFICE SPARGER

Various arrangements of perforated pipes have been tried in different types of fermentation vessel with or without impellers. In small stirred fermenters the perforated pipes were arranged below the impeller in the form of crosses or rings (ring sparger), approximately three-quarters of the impeller diameter. In most designs the air holes were drilled on the under surfaces of the tubes making up the ring or cross. Walker and Holdsworth (1958) commented that in production vessels, sparger holes should be at least 6 mm (1/4 inch) diameter because of the tendency of smaller holes to block and to minimize the pressure drop.

In low viscosity fermentations sparged at 1 vvm (volume of air^{-1} volume of medium^{-1} minute^{-1}) with a power input of 1 W kg^{-1}, Nienow *et al.* (1988) found that the power often falls to below 50% of its unaerated value when using a single Rushton disc turbine which is one-third the diameter of the vessel and a ring sparger smaller than the diameter of the agitator. If the ring sparger were placed close to the disc turbine and its diameter was 1.2 times that of the disc turbine, a number of benefits could be obtained (Nienow *et al.*, 1988). A 50% higher aeration rate could be obtained before flooding occurred, the power drawn was 75% of the unaerated value, and a higher $K_L a$ could be obtained at the same agitator speed and aeration rate. These advantages were lost at viscosities of about 100 m Pas.

Orifice spargers without agitation have been used to a limited extent in yeast manufacture (Thaysen, 1945), effluent treatment (Abson and Todhunter, 1967) and later in the production of single-cell protein in the air-lift fermenter which are discussed in a later section of this chapter (Taylor and Senior, 1978; Smith, 1980).